Consumer Valuation of Battery Health Attributes in Used Battery Electric Vehicle Markets: A Discrete Choice Experiment

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ARTICLE INFO

Keywords:
Battery electric vehicle
Used vehicle market
Discrete choice experiment
Willingness-to-pay
Consumer preferences
Information treatment

ABSTRACT

Buyers of used battery electric vehicles (BEVs) face substantial uncertainty about battery condition at the point of sale, even though the battery accounts for much of the vehicle's value. This study delivers the first consumer willingness to pay (WTP) estimates for battery-health attributes of used BEVs . A discrete choice experiment was administered to a national sample of 3,072 used vehicle intending buyers in the U.S. The DCE jointly varies price, mileage, electric range, battery state of health,and battery refurbishment history across six repeated choice tasks. A mixed logit model shows that A mixed logit model shows that consumers value additional range at $11,070 per 100 miles, discount mileage at $2,780 per 10,000 miles and battery degradation at roughly $980 per percentage point of annual range loss, and penalize pack- and cell-level refurbishment by $4,060 and $4,480. Each attribute shows wide dispersion around its mean, suggesting substantial heterogeneity in their WTP across the population. A latent class choice model further identifies six consumer segments, from *EV-engaged, range-focused* group to *battery-quality-attentive* plurality to *budget-constrained* group to *BEV skeptics*. These classes differ markedly in their sensitivity to battery-health attributes, price, and the no-choice option. Attitudes, infrastructure access, and demographics explain much of the heterogeneity. In addition, respondents who reject all vehicle offers most often cite *general BEV disinterest*, *economic barriers*, and *charging inconvenience*. However, providing detailed battery replacement cost and warranty information does not affect overall market engagement and choice behavior. This study assesses the potential for battery-health disclosure to serve as a market-transparency signal in used BEV transactions, which carries implications for used vehicle dealers, OEMs, and certification service providers.

1. Introduction

The used vehicle market is central to U.S. automobile ownership: used vehicle transactions account for roughly 70% of all passenger vehicle purchases and are projected to grow steadily through the coming decade (Market Research Future, 2025). An increasing number of used battery electric vehicles (BEVs) are entering the secondary market and the demand is also accelerating: sales rose 34.2% year-over-year through early 2025, while average days of supply fell by 21.5% (Cox Automotive Inc., 2025). The growth of the used BEV market extends EV access to middle- and lower-income households (Hagman et al., 2016), can support a circular economy by extending vehicle and battery lifetimes (Guzek et al., 2024), and underpins the residual values that make new EV leases financially viable (Brückmann et al., 2021; Lim et al., 2015).

Despite this momentum, consumer hesitation toward used BEVs remains pronounced. Fernandes (2023) reports that 52% of U.S. adults are unlikely to consider purchasing a used EV, compared to 28% who express openness to the idea. In the same survey, consumers most often cite *uncertainty about vehicle and battery condition* (41%) and *concerns about reliability relative to new EVs* (23%) as primary barriers. Among those who have already purchased a used EV, over half reported pre-purchase concerns about battery longevity, even though only 5.1% experienced

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substantial performance loss in practice (Sheykhfard et al., 2025). This gap between perceived and actual risk points to an information problem: buyers lack information about battery health at the point of sale, even though the battery can account for up to 30% of a used BEV's value (Boudway, 2020).

Unlike conventional vehicles, where odometer reading and vehicle age are reasonable proxies for overall wear, battery degradation in BEVs depends on factors such as charging history, depth of discharge cycles, thermal exposure, and driving patterns (Bashash et al., 2011; Neubauer et al., 2012). Two BEVs of identical age and mileage can differ substantially in their remaining battery capacity and projected useful life. Little of this history is visible to buyers, and some of it is unavailable even to sellers themselves. As a result, sellers cannot credibly signal battery quality and struggle to recoup the true value of their vehicles at resale, while buyers, unable to tell the difference, may undervalue well-maintained EVs and overpay for poorly maintained ones. The consequences extend beyond individual transactions: persistent market uncertainty may suppress new BEV adoption by generating "resale anxiety" among prospective buyers who question whether their new vehicle will hold its value (Brückmann et al., 2021).

Webb et al. (2025) finds that both BEV current owners and potential buyers view battery information, including expected lifetime and range, as very useful for purchasing decisions. Stated usefulness, however, does not measure consumers' willingness to pay for these attributes. The study measures vehicle buyers' preferences by administering a discrete choice experiment (DCE) to a national sample of 3,072 U.S. adults who reported an intention to purchase a used car or SUV within the next two years. The DCE presents respondents with six repeated choices among hypothetical used BEVs that vary in mileage, purchase price, electric range, battery degradation rate, and battery refurbishment history. Respondents are randomly assigned to receive either basic range-and-health information or extended information that additionally describes battery maintenance costs and warranty coverage. This allows us to test whether the information treatment shapes preferences and opt-out behavior. Respondents who selected the "no-choice" option in all six choice tasks were asked to describe, in their own words, the reasons for consistently opting out.

This study provides the first WTP estimates for battery-health attributes in the U.S. used BEV market, *conditional on attributes being visible to consumers*. We estimated two complementary models: (1) a mixed logit (MXL) model characterizes aggregate preferences and the degree of preference heterogeneity across the full sample, and (2) a latent class choice model (LCCM) then partitions respondents into six distinctive market segments that differ in their sensitivity to vehicle attributes and individual characteristics. The MXL results suggest that the population, on average, values additional electric range ($11,070 per 100 miles at Year 3), penalizes higher mileage ($2,780 per 10,000 miles), and discounts battery degradation rate (approximately $980 per percentage point of annual range loss). Refurbishment history carries significant disutility, though large standard deviation estimates indicate substantial variation across individuals. The LCCM indicates that this heterogeneity has a discrete structure. Approximately 17% of respondents belong to an EV-engaged segment that places a high premium on driving range, while an additional 45% fall into two classes that are particularly attentive to battery health attributes. A further 17% are either price-constrained buyers or deeply skeptical non-adopters unlikely to enter the used BEV market in the near term. The presence of identifiable segments with strong preferences for range and battery health supports the potential for premium pricing for used BEVs with certified battery health, whereas the price-sensitive segment suggests continued demand for affordable used BEVs. We also conduct a thematic analysis that identifies 10 barriers to BEV market disengagement, with the most frequently cited being *general BEV disinterest*, *economic barriers*, and *charging inconvenience*. Finally, we find that the information treatment does not significantly affect the overall rate of market disengagement.

2. Literature Review

This section reviews four strands of literature that inform the survey design and analysis: used-EV adoption, battery degradation and state of health (SOH), replacement cost and warranty considerations, and consumer valuation of battery-health attributes.

2.1. Used Electric Vehicle Adoption

Prior reviews organize the drivers of and barriers to EV adoption into several domains: perceptions and attitudes, sociodemographic characteristics, technology reliability, infrastructure accessibility, economic constraints, peer and community influences, and information and knowledge (Gore et al., 2025; Ioganseen et al., 2026, 2023; Lohawala and Rahman, 2026; Naseri et al., 2024; Sonar et al., 2023; Yuan et al., 2018; Zou et al., 2020). Most of these studies concern the new-EV market, whereas fewer studies examine the used market, even as it becomes an increasingly important channel into EV ownership.

While used-EV buyers share some attribute preferences with new-EV buyers, the two consumer segments can differ in profile, decision context, and constraints. Demographically, studies find that used-EV adopters are more often male, White, better educated, and more likely to live in detached homes or townhouses, yet report lower household incomes than new-EV buyers. This suggests that used EVs offer a cheap entry into EV ownership (Mashrur and Mohamed, 2025; Sheykhfard et al., 2025; Tal et al., 2017). Structurally, used-EV adopters more often lack on-site charging: in a stated-preference experiment, used-car buyers responded most to the proximity of slow charging near home ("garage orphans"), whereas new-EV buyers responded to charging availability at their home parking space (Zou et al., 2020). Affordability and battery condition also constrain used-EV buyers in ways that constrain new-EV buyers less (Canepa et al., 2019; Loh and Noland, 2024). In terms of daily travel patterns, range, though a common EV concern, may be less binding in the used market: Sheykhfard et al. (2025) found that 59.4% of used EV owners drove less than 100 miles per day and another 29.4% drove 101–200 miles, distances well within the range of most current used BEVs.

2.2. Battery Degradation and State-of-Health

The uncertainty specific to used BEVs is battery health, the remaining useful life of the battery (Pedrosa and Nobre, 2018). What buyers cannot observe is battery degradation, the loss of capacity and driving range over time, which depends on how frequently and deeply the battery is cycled, the state-of-charge windows at which it is stored, and the ambient temperatures it is exposed to (Bashash et al., 2011; Neubauer et al., 2012). Yang et al. (2018) simulate battery useful life across U.S. driving and climate conditions and find that expected lifetimes ranging from roughly five years for hot-climate, high-mileage owners to thirteen years for moderate-climate light users. Battery typically degrade linearly or sublinearly but can acceleration later in life, making long-term condition difficult to predict (Attia et al., 2022). The conventional retirement benchmark, an SOH of 70–80% range relative to when the pack was new (Canals Casals et al., 2019), is a rule of thumb rather than a hard physical threshold: the actual retirement point depends on the vehicle segment, warranty policy, and the available second-life pathway.

For used-BEV buyers, the two quantities most relevant to valuation (how much life remains in the pack, and how confidently that estimate can be trusted) are precisely the hardest to observe. Even when SOH is reported, the number itself is not standardized. Neither industry nor academia has reached consensus on a vehicle-level SOH definition or measurement procedure Bilfinger et al. (2025), and manufacturer-reported SOH can be inconsistent and unreliable across EV platforms Park et al. (2026). This problem is especially acute in secondary markets, where buyers lack knowledge of prior use, charging behavior, and refurbishment history. Battery-health disclosure is therefore a market-transparency problem as much as a technical one.

2.3. Battery Replacement, Cost, and Warranty Considerations

Battery warranty coverage and out-of-warranty replacement costs anchor the economics of the used-BEV market because they shape both expected ownership costs and perceived downside risk. Manufacturer battery warranties are relatively standardized, typically eight years or 100,000 miles (whichever comes first), with replacement triggered at some manufacturers when capacity falls below a stated threshold (commonly 70%) (Clarke, 2024). Empirically, average degradation stays well within these bounds. Drawing on a fleet of roughly 5,000 EVs, Dnistran (2024) reports a mean annual capacity loss of only 1.8%, implying that under typical duty cycles, the pack should outlast the vehicle's expected service life. Even so, out-of-warranty replacements, when they occur, commonly cost \$5,000 to \$16,000 and exceed \$20,000 for some models, depending on pack size, labor, and parts availability (Kothari, 2024; Witt, 2024), enough to keep replacement risk salient for prospective buyers. total-cost-of-ownership (TCO) studies treat battery lifetime, replacement cost, and resale value as joint determinants of BEV ownership costs (Hagman et al., 2016; Letmathe and Suares, 2017), but because these attributes are hard to verify at the point of sale, they stop short of estimating how used-BEV buyers actually price battery-condition uncertainty.

As growing numbers of EV batteries reach the end of their vehicle life, their disposition has become a live commercial and policy challenge (Skeete et al., 2020; Tankou et al., 2023). The industry has largely converged on two routes: repurposing low-capacity batteries for stationary energy storage (the "second-life" route), or restoring batteries for continued use in vehicles through refurbishment. Refurbishment can recover residual value from batteries after their first vehicle life and may make refurbished options more cost-competitive than new packs (Jiao and Evans, 2016; Shaikh et al., 2023). Studies of remanufactured products suggest that consumers often discount refurbished goods because of perceived quality risks and uncertainty about the refurbishment process (Abbey et al., 2015). Whether these concerns extend to refurbished EV batteries remains largely unexplored. Pedrosa and Nobre (2018) is the only studies to examine consumer perceptions of used EVs with replaced batteries. Their qualitative results suggest that pairing a replacement battery with an explicit dealer warranty improves sentiment and, for some buyers, raises reported WTP above the baseline used-EV scenario. Because that study is qualitative, small in scope, and conducted in Portugal, its findings may not generalize to the U.S. market. A further question concerns the depth of refurbishment, since batteries can be disassembled and repaired at the pack, module, or cell level. Foster et al. (2014) estimate that remanufactured batteries with replaced cells or modules cost about 40% less than newly manufactured packs. However, these savings carry technical risk: pairing new modules with aged ones can unbalance the pack, and opening a sealed pack makes the repairer responsible for restoring its sealing, thermal management, and electrical isolation (Tire Review Staff, 2026). Whether consumers evaluating used BEVs differentiate between original batteries and refurbished batteries in different levels (e.g., cell- or module-level replacement versus complete pack replacement) remains an open empirical question.

2.4. Willingness to Pay for Vehicle and Battery-Health Attributes

Direct WTP evidence on vehicle and battery-health attributes of used EVs remains limited. A substantial literature, however, focuses on the attributes of new EVs. Table 9 in the Appendix consolidates the monetized estimates from these studies by attribute.

Driving range is the most studied attribute. Hidrue et al. (2011), using a choice experiment amaong potential BEV buyers in the U.S., estimate WTP of $35–$75 per additional mile of range. A meta-analysis of 33 stated-preference studies on BEVs and broader alternative fuel vehicles places mean WTP at US$66–$75 per mile and finds that marginal WTP declines as baseline range grows, so additional range is worth most for short-range vehicles (Dimitropoulos et al., 2013). Tanaka et al. (2014) report a far lower value for BEV and plug-in EV (PHEV) choices ($21.5 per 10 miles) from a design whose range levels extended to 1,000 miles. The most recent U.S. benchmark is Forsythe et al. (2023): in a nationally representative DCE of new-BEV buyers, car and SUV buyers are willing to pay $5,120 and $7,010, respectively, per 100 miles of additional BEV range.

Consumers also price other usability and performance attributes. Estimated WTP includes $425–$3,250 per hour reduction in charging time (Hidrue et al., 2011), and roughly $4,100 for fast-charging capability (Forsythe et al., 2023). A $0.01-per-mile reduction in operating cost is worth about $1,600 in the U.S. (Helveston et al., 2015), or $1,960 for cars and $1,490 for SUVs in Forsythe et al. (2023), and a one-second acceleration improvement is worth roughly $1,000–$1,500 across U.S. studies (Forsythe et al., 2023; Greene et al., 2018; Helveston et al., 2015). Helveston et al. (2015) further estimate that U.S. consumers value BEV technology $10,000–$20,000 below a comparable conventional vehicle depending on range. Pooling 52 U.S. studies and 777 marginal WTP estimates across 142 attributes, Greene et al. (2018) report mean values of $86 per mile of range, $2,195 per hour of recharging time saved, and $1,880 per one-cent-per-mile fuel-cost reduction. Vehicle attributes, in short, are routinely capitalized into consumer WTP.

Evidence on battery condition, the attributes closest to our study, is far thinner. In a Canadian national latent class study among EV potential buyers, Ferguson et al. (2018) report WTP of up to $3,153 for a one-level improvement in battery warranty, alongside about $30 per kilometer of added range among PHEV-oriented classes, $1,971 per hour of public charging time saved, and $1,122 for a step improvement in charging-station availability.

Two features of this literature matter for interpreting our results. First, all estimates are nominal in each study's survey year, so cross-study comparisons require inflation adjustment. Second, no existing study prices the attributes that distinguish a used BEV: mileage, SOH and degradation rate, and refurbishment history. The present study addresses this gap.

3. Survey Design, Data Collection, and Sample Profile

3.1. Nationwide Used EV Survey

The survey elicits consumer preferences and WTP for vehicle and battery-health attributes in the used BEV market. The target population is U.S. adults aged 18 or older who reported their intention to purchase a used car or SUV within the next two years. For a full description of the survey instrument, see Gore et al. (2025).[7]

The centerpiece of the survey is a BEV DCE, designed to examine how battery-health attributes are valued. The DCE includes six repeated choice tasks in which respondents choose among hypothetical used BEVs with varying attributes. All vehicles are assumed to have been manufactured in 2022 and to be three years old at the time of purchase. The attributes varied are vehicle mileage, purchase price, electric range, battery degradation and SOH, and battery refurbishment history. Before the choice tasks, respondents see descriptions of all attributes, including definitions for

[7]The Data Collection Instrument was published prior to data collection. Certain aspects of the survey design were revised following piloting. Where discrepancies exist between the instrument and this paper, the description in this paper reflects the final implemented version.

battery-specific terms (see Table 1). The DCE design varied two underlying parameters: electric range at manufacture (Year 0) and annual degradation rate. To improve comprehension, respondents instead saw these transformed into range and SOH, both at Year 3 (the point of purchase) and Year 8 (five years after purchase).

The DCE incorporates several ways to increase contextual tangibility and behavioral realism (Haghani et al., 2021). First, each respondent selected a vehicle image of their prefered body type (car versus SUV) they found appealing before the choice tasks. This image was then used to depict every alternative across the six tasks, holding appearance constant so that choices reflect the listed attributes rather than styling. Second, for choice task, respondents select among three used BEVs or a “None of the above” option. The opt-out allows respondents to decline unattractive alternatives, which makes the stated choices closer to actual purchase decisions. Third,respondents were assigned to one of four DCE versions based on their stated budget (low, ≤$20K versus high,>$20K) and body-type preference.

Choice profiles were generated as a random design using the `cbcTools` R package (Helveston, 2025): for each respondent, attribute levels for each alternative were drawn at random from the level ranges of the assigned version. Unlike efficiency-optimized designs, a random design requires no prior assumptions about preference parameters and preserves the independent attribute variation needed to identify interaction and nonlinear effects. Figure 1 shows an example.

Respondents were also randomly assigned to one of two information treatment conditions. The *basic information* group received descriptions of electric range and battery SOH only. The *extended information* group additionally received information on expected battery useful lifetime, replacement cost, and warranty coverage. This design allows us to test whether the content of battery information affects consumers’ choices and their propensity to opt out.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget (≤$20K) Car	**High-budget** (>$20K) Car	**Low-budget** (≤$20K) SUV	**High-budget** (>$20K) SUV
Mileage	The total number of miles a vehicle has traveled since manufacture.	15,000–60,000 miles, in increments of 500 miles.			
Battery refurbishment history	The repair or replacement work that has been performed on the vehicle’s main battery pack.	*(1) Original:* The battery remains in its factory-original condition with no repairs or replacements. *(2) Some battery cells replaced:* A portion of the battery cells has been replaced to address performance issues or extend battery life; the remaining cells are original. *(3) Entire battery pack replaced:* The full battery pack has been replaced with a new, refurbished, or used pack, typically after battery performance declined below a specified threshold.			
Electric range at delivery (Year 0)	The maximum distance the vehicle can travel on a full battery charge under typical driving conditions, as rated at the time of manufacture.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Battery annual degradation rate and state of health (SOH)	The average annual percentage loss in electric range and battery SOH. The remaining range and SOH at Year 3 (current) and Year 8 (five years from purchase) are derived from this rate.	1%–8%, in increments of 1%.			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	$10,000–$20,000 ($2,000 increments)	$20,000–$40,000 ($5,000 increments)	$15,000–$25,000 ($2,000 increments)	$25,000–$45,000 ($5,000 increments)

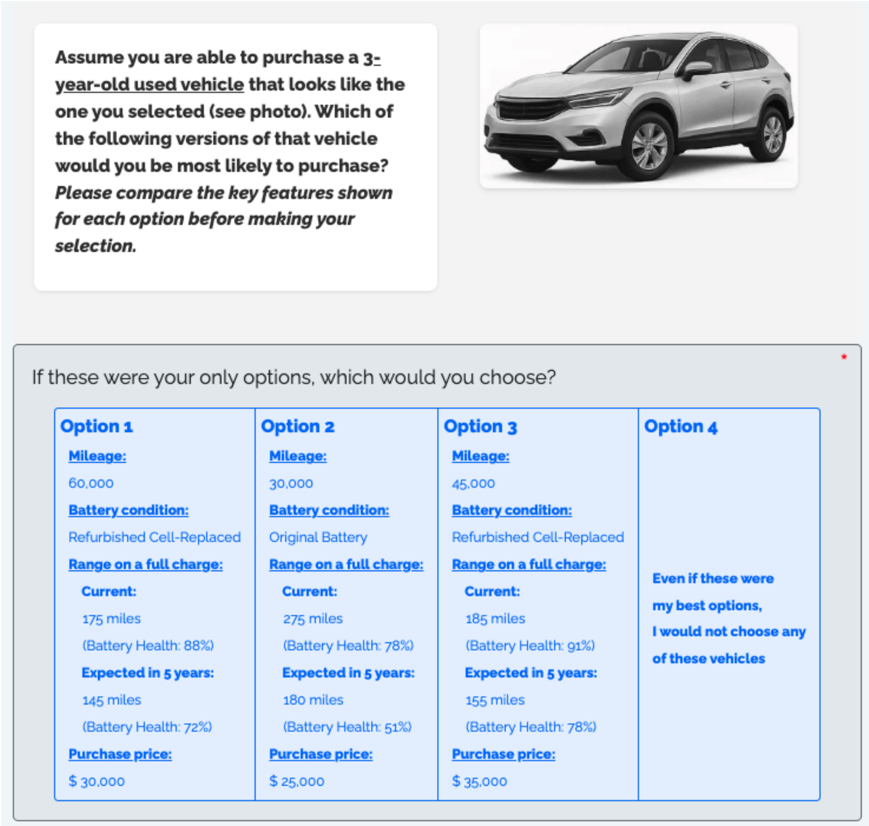

Figure 1: Example of a choice experiment.

In addition to the DCE, the survey collects additional information. Before the choice tasks, respondents report on their current household vehicle fleet (fuel type, range, acquisition method, and costs) and future purchasing plans (anticipated budget, payment method, and the likelihood of buying a new or used plug-in hybrid or BEV). After the choice tasks, the survey assesses respondents' knowledge, attitudes, and perceptions of EVs and battery technology. This section includes factual questions on EV charging capabilities and federal tax incentives, as well as psychological statements rated on five-point Likert scales, covering social norms, environmental beliefs, cost perceptions, battery concerns, and personal traits such as risk-taking and price sensitivity. Two items about refurbished EV batteries are of particular relevance to this study: "Purchasing refurbished EV batteries will minimize negative effects on natural ecosystems" (hereafter *environmentally positive*) and "Refurbished EV batteries do NOT perform and function as original EV batteries" (hereafter *functionally negative*). The final section collects demographic information including age, gender, race and ethnicity, household size, income, education, employment status, and housing type.

3.2. Data Collection and Sample Profile

The survey was administered online through two research platforms (Qualtrics and Dynata) between December 2025 and April 2026. The median completion time was approximately 14 minutes. All participants received platform-standard compensation.

The initial sample comprised 3,657 respondents from across the U.S. We identified and removed low-quality responses by reviewing each case, including overall completion-time speeding, failure of the embedded attention check, internal inconsistencies across survey sections, and uninformative or incoherent open-ended text. A separate timing screen flagged respondents who sped through the choice tasks specifically.

The final sample consists of 3,072 respondents, yielding 18,432 choice observations across six tasks. Figure 2 shows the geographic distribution of respondents by ZIP code, with point size and color intensity scaled to respondent count.

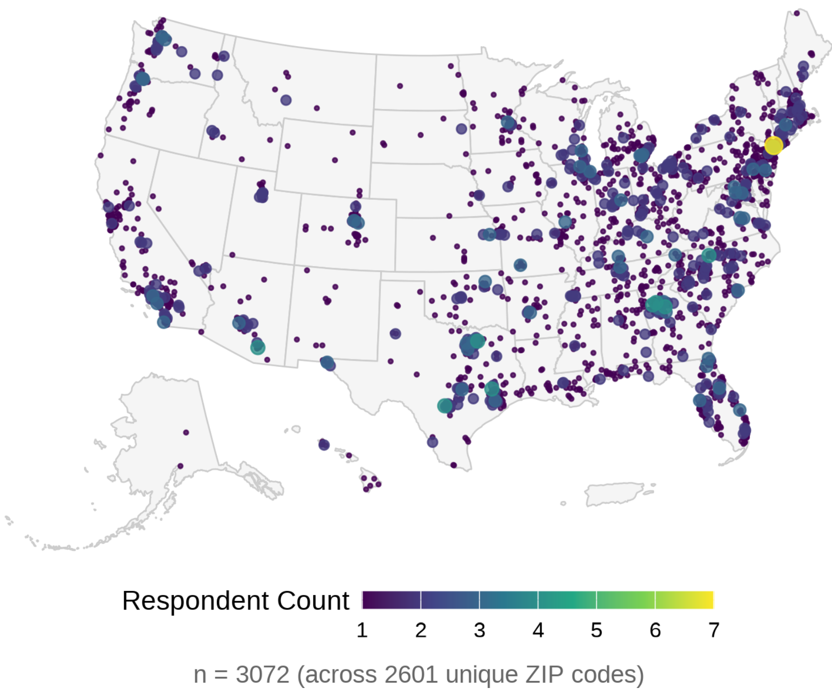

Figure 2: Geographic distribution of survey respondents across the continental United States.

Because no official population distribution exists for consumers intending to purchase a used vehicle, we compare the sample's demographic profile with ACS 2020–2024 five-year estimates (Table 2). The sample aligns closely with the national adult population on housing type and on lower- and middle-income shares. It skews younger (5.4% aged 65 and older versus 22.0% nationally), more female (60.7% versus 51.0%), and more educated (54.1% hold a bachelor's degree or higher versus 35.7% of adults 25 and older). Renters are over-represented (46.7% versus 34.8%), while households earning $150,000 or more (13.4% versus 23.0%) and Hispanic adults (11.4% versus 17.4%) are under-represented. Some of these deviations are consistent with findings from prior studies of used-vehicle buyers (e.g., skew younger and lower-income than the general population). Without the population benchmark, we decided not to weight the data, and all analyses below are unweighted.

Table 2
Sample profile compared with the U.S. population (ACS 2020–2024 5-year estimates).

	Sample (n)	Sample (%)	ACS (%)
Age			
18 - 24	438	14.4	11.7
25 - 34	921	30.2	17.5
35 - 44	742	24.4	16.9
45 - 54	500	16.4	15.6
55 - 64	280	9.2	16.2
65+	165	5.4	22.0
Gender			
Female	1,861	60.7	51.0
Male	1,140	37.2	49.0
Other	65	2.1	–
Race			
White alone	2,088	68.0	63.3
Black or African American alone	478	15.6	11.9
Other or multiple races	506	16.5	24.8
Hispanic origin			
Hispanic or Latino	349	11.4	17.4
Not Hispanic or Latino	2,723	88.6	82.6
Household income			
Less than $25,000	458	14.9	14.6
$25,000 - $49,999	495	16.1	16.7
$50,000 - $74,999	751	24.5	15.5
$75,000 - $99,999	354	11.5	12.6
$100,000 - $149,999	601	19.6	17.5
$150,000 or more	411	13.4	23.0
Education			
High school or less	439	14.3	36.4
Some college or associate	968	31.6	27.9
Bachelor's degree	1,065	34.8	21.6
Graduate or professional degree	592	19.3	14.1
Housing tenure			
Own	1,519	53.3	65.2
Rent	1,331	46.7	34.8
Housing type			
Single-family detached	1,874	61.0	62.3
Single-family attached	330	10.7	6.4
Apartment (2+ units)	755	24.6	26.0
Mobile home or other	113	3.7	5.2

Notes: Sample size and shares are unweighted and exclude item nonresponse. '–' indicates no comparable ACS category. ACS universes: population 18+ (age, gender, race, Hispanic origin), population 25+ (education), and households (income, tenure, housing type).

4. Method

4.1. Mixed Logit Model

We begin with estimating WTP for vehicle attributes across the full sample (3,072 respondents, 18,432 choice observations). We estimate a mixed logit (MXL) model directly in *WTP space* using the *logitr* R package (Helveston, 2023). This provides an aggregate baseline and reveals the extent of preference heterogeneity. The MXL model relaxes the Independence of Irrelevant Alternatives (IIA) assumption of the standard multinomial logit by allowing preference parameters to vary continuously across individuals (McFadden and Train, 2000; Train, 2009). Estimating in WTP space lets us specify the distribution of WTP for each non-price attribute directly, rather than recovering it post hoc from the ratio of attribute to price coefficients. We assume individual WTP values follow normal distributions, which captures symmetric heterogeneity around population means. The utility parameterization follows the notation in Helveston et al. (2018) and Helveston (2023).

The WTP-space utility for respondent n choosing alternative i in choice scenario t is:

$$u_{nti} = \lambda_n\left(\boldsymbol{\omega}_n^\top \mathbf{x}_{nti} - p_{nti}\right) + \varepsilon_{nti}, \qquad \varepsilon_{nti} \sim \text{Gumbel}(0,1) \tag{1}$$

where p_{nti} is the purchase price presented in scenario t to respondent n; $\mathbf{x}_{nti}$ is a vector of non-price BEV attributes including vehicle mileage, battery refurbishment history, current electric range at the point of purchase (Year 3), and the proportional rate of range loss over the subsequent five years (to Year 8);[8] $\lambda_n > 0$ is the individual-specific scale parameter; $\boldsymbol{\omega}_n$ is a vector of individual WTP values drawn from a mixing distribution $f(\boldsymbol{\omega} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ with population mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$; and ε_{nti} is an IID Type I Extreme Value error. Because λ_n and $\boldsymbol{\omega}_n$ are both unobserved, the choice probability requires integrating the logit kernel over their joint mixing distribution:

$$P_{nti}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \int\!\!\int \frac{\exp\left(\lambda\left(\boldsymbol{\omega}^\top \mathbf{x}_{nti} - p_{nti}\right)\right)}{\sum_j \exp\left(\lambda\left(\boldsymbol{\omega}^\top \mathbf{x}_{ntj} - p_{ntj}\right)\right)} f(\boldsymbol{\omega} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})\, g(\lambda)\, d\boldsymbol{\omega}\, d\lambda \tag{2}$$

This integral is evaluated by simulation using Halton draws ($R = 1000$). Accounting for the panel structure of six repeated choice tasks per respondent, the simulated log-likelihood is:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln\left[\frac{1}{R}\sum_{r=1}^{R}\prod_{t=1}^{T} \frac{\exp\left(\lambda_r\left(\boldsymbol{\omega}_r^\top \mathbf{x}_{nti^*_{nt}} - p_{nti^*_{nt}}\right)\right)}{\sum_j \exp\left(\lambda_r\left(\boldsymbol{\omega}_r^\top \mathbf{x}_{ntj} - p_{ntj}\right)\right)}\right] \tag{3}$$

where $(\lambda_r, \boldsymbol{\omega}_r)$ is the r-th draw from the joint mixing distribution and i^*_{nt} is the alternative chosen by respondent n in scenario t. Because estimation is carried out directly in WTP space, the population mean WTP for each non-price attribute k is obtained directly as $\hat{\mu}_k$.

4.2. Latent Class Choice Model

Model Framework

The MXL results reveal substantial preference heterogeneity across respondents. To explore whether this heterogeneity reflects discrete consumer segments with systematically different preferences, we estimate a latent class choice model (LCCM) using the *Apollo* R package. The LCCM assumes that individuals can be assigned to a finite number of unobserved groups (classes) within which preferences are homogeneous, but across which they differ (Greene and Hensher, 2003; Kamakura and Russell, 1989). This structure is well suited to the situations where consumers may differ in their attitudes toward battery health risk, their familiarity with EV technology, and their price sensitivity. The model is estimated on a subsample obtained after excluding cases with missing data on covariates (2,916 respondents, 17,496 choice observations).

As illustrated in Figure 3, the LCCM consists of two simultaneously estimated components. The *class membership model* assigns each respondent probabilistically to a latent class as a function of observable individual characteristics. The *class-specific choice model* then estimates the conditional probability of each vehicle choice within each class. In addition, a set of *inactive indicators* excluded from estimation for collinearity or theoretical reasons is used to characterize the classes descriptively after estimation. The formal model structure follows the notation of Chen et al. (2023).

[8] We compared six attribute specifications that varied whether range was expressed at Year 0 or Year 3, and whether battery health change was expressed as absolute range loss or as a percentage degradation rate. The specification combining range at Year 3 with percentage loss rate from Year 3 to Year 8 yielded the best model fit and the most economically interpretable coefficients, which suggests consumers evaluate used BEVs by current performance and relative future deterioration.

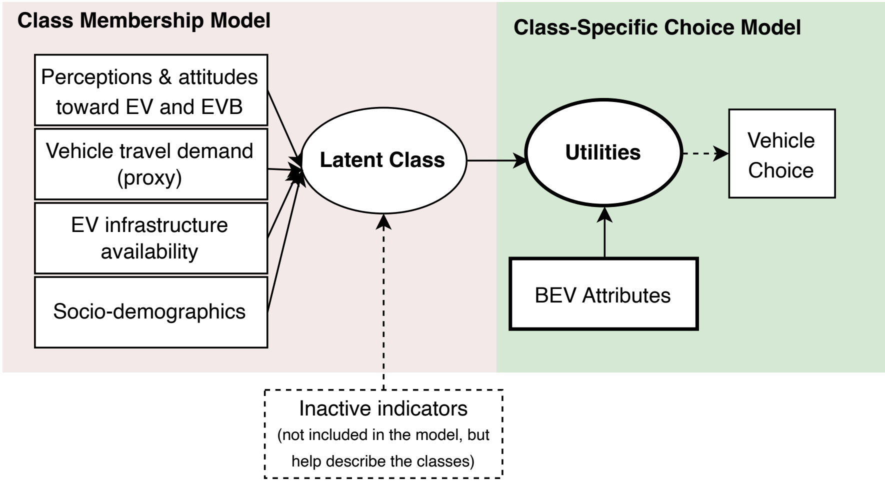

Figure 3: LCCM model framework.

Class-specific Choice Model

For respondent n belonging to latent class s ($s = 1, 2, \dots, S$), the utility of choosing vehicle alternative i in choice scenario t ($t = 1, \dots, 6$) is:

$$U_{nti|s} = \boldsymbol{\beta}_s^\top \mathbf{X}_{nti} + \varepsilon_{nti|s} \tag{4}$$

where $\mathbf{X}_{nti}$ is a vector of observed BEV attributes (vehicle mileage, battery refurbishment status, electric range at Year 3, proportional range loss to Year 8, and purchase price); $\boldsymbol{\beta}_s$ is a vector of class-specific utility parameters; and $\varepsilon_{nti|s}$ is an IID Type I Extreme Value error. Under this assumption, the conditional probability that respondent n in class s chooses alternative i in scenario t follows the MNL form:

$$P_{nt}(i \mid s) = \frac{\exp\left(\boldsymbol{\beta}_s^\top \mathbf{X}_{nti}\right)}{\sum_{j \in C_{nt}} \exp\left(\boldsymbol{\beta}_s^\top \mathbf{X}_{ntj}\right)} \tag{5}$$

where C_{nt} denotes the choice set available to respondent n at scenario t.

To capture the non-linear relationship between range and choice probability, we adopt a piecewise linear specification for the range and degradation rate attributes. Consumers likely have diminishing sensitivity to range at higher levels and heightened sensitivity around near minimum viability thresholds (Dimitropoulos et al., 2013; Hackbarth and Madlener, 2016). We compared a quadratic specification, which imposes a symmetric parabolic shape and performs poorly at the tails, with a piecewise linear specification that estimates separate slopes for distinct segments. The piecewise specification provides superior model fit and more interpretable coefficients. Electric range at Year 3 is segmented into 40–130 miles, 130–200 miles, and above 200 miles; the range loss rate is segmented into below 12%, 12–24%, and above 24%. These thresholds reflect both theoretical considerations about minimum driving viability and empirical patterns in the data (the breakpoints fall near the 33rd and 66th percentiles of each attribute's distribution). To assess whether the estimated marginal valuations differ significantly across segments, we apply the delta method to test pairwise WTP differences within each class. For any two segments s and s', the WTP difference is $(\hat{\beta}_s - \hat{\beta}_{s'})/(-\hat{\beta}_{\text{price}}) \times 10{,}000$, with standard errors approximated using the model's robust covariance matrix. Full results are reported in Appendix Table 8. Note that pairwise differences are meaningful only when the individual WTP estimates being compared are estimated with sufficient precision. When either the attribute or price coefficient is not statistically distinguishable from zero, the resulting WTP estimate is imprecise, and the pairwise comparison inherits this uncertainty. Consequently, a non-significant pairwise difference may indicate either genuinely similar marginal valuations or insufficient statistical power to distinguish two imprecisely estimated WTPs. For this reason, we discuss pairwise differences only for classes in which the underlying attribute-specific WTP estimates are themselves statistically significant.

Class Membership Model

The class membership model specifies the probability that respondent n belongs to latent class s as a function of observed individual characteristics:

$$M_n(s) = \frac{\exp\left(\boldsymbol{\gamma}_s^\top \mathbf{Z}_n\right)}{\sum_{s'=1}^{S} \exp\left(\boldsymbol{\gamma}_{s'}^\top \mathbf{Z}_n\right)} \tag{6}$$

where $\mathbf{Z}_n$ is a vector of individual-level covariates organized into four groups: (1) *perceptions and attitudes toward EVs and EV batteries* (e.g., range anxiety, environmental and functional assessments of refurbished batteries); (2) *vehicle travel demand proxies* (e.g., range of the respondent's primary vehicle); (3) *EV infrastructure availability* (e.g., access to electrical outlets for charging); and (4) *socio-demographic characteristics* (e.g., household income). The parameters for the first class ($s = 1$), which serves as the reference class, are normalized to zero for identification.

The unconditional probability that respondent n chooses alternative i in scenario t, integrating over all latent classes, is:

$$P_{nt}(y_n = i) = \sum_{s=1}^{S} P_{nt}(i \mid s) \cdot M_n(s) \tag{7}$$

Because each respondent completes all $T = 6$ choice tasks as a member of a single latent class, the likelihood takes the product over tasks within each class before summing across classes. The log-likelihood across all N respondents is:

$$\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \sum_{n=1}^{N} \ln \left[\sum_{s=1}^{S} M_n(s) \prod_{t=1}^{T} P_{nt}(i_{nt}^* \mid s) \right] \tag{8}$$

where i_{nt}^* denotes the alternative chosen by respondent n in scenario t. The model parameters $\boldsymbol{\beta}_s$ and $\boldsymbol{\gamma}_s$ are estimated jointly via maximum likelihood.

Willingness to Pay

A key output of the LCCM is class-specific WTP for each BEV attribute. Because price enters linearly, the marginal utility of income is constant within each class, and the WTP for attribute k in class s is:

$$\text{WTP}_{s,k} = -\frac{\hat{\beta}_{s,k}}{\hat{\beta}_{s,\text{price}}} \tag{9}$$

This represents the dollar amount a representative respondent in class s would pay, or would need to be compensated, for a one-unit change in attribute k, holding all other attributes constant.

4.3. Thematic Analysis of Reasons for Opt-out

Respondents ($n = 209$) who opted out systematically (i.e., choosing none of the presented used BEVs in all six choice tasks) were asked to explain their reasons in an open-ended question at the end of the DCE. These open-ended responses complement the quantitative analyses in two ways: they point to the specific attributes respondents dislike most, and they can surface barriers that are absent from our attribute list or hard to quantify in closed-ended questions.

We use thematic analysis, which applies a coding system to organize qualitative responses and identify recurring themes (Braun and Clarke, 2006). Reviewing the response texts, we constructed a two-level codebook of 10 parent themes and 25 sub-themes. Because respondents could cite multiple reasons for opting out, we also examine theme co-occurrence to identify clusters of related barriers (Krishna, 2021). Finally, we test whether the battery information treatment affects the opt-out rate.

5. Results and Discussion

5.1. Results of Mixed Logit Model

Table 3 presents the MXL estimates. All mean WTP parameters are significant at the 0.1% level (***). On average, respondents value an additional 100 miles of electric range at Year 3 at $11,070, penalize every 10,000 miles of odometer reading by $2,780, and discount each additional percentage point of annual range-loss rate by approximately $980. Both refurbishment types generate meaningful disutility: pack replacement reduces WTP by $4,060 and cell replacement by $4,480. The no-choice opt-out option carries a large negative WTP (-$51,300), reflecting a strong baseline preference to purchase a vehicle rather than opt out.

The standard deviation estimates show substantial preference heterogeneity across respondents. Range at Year 3 exhibits the widest dispersion in absolute terms (SD = $13,950 relative to a mean of $11,070), with the distribution of simulated WTP draws spanning from $1,660 at the first quartile to $20,470 at the third quartile. This indicates that while most respondents positively value range, a non-trivial share places little weight on it. Heterogeneity in the refurbishment parameters is also pronounced: the standard deviations for pack replacement ($7,300) and cell replacement ($6,720) are both larger in magnitude than their respective means, and the upper quartiles of the simulated draws are positive ($850 and $70, respectively). This implies that roughly one quarter of consumers view battery refurbishment neutrally or even as a positive quality signal, potentially because a replaced battery is perceived as newer or more cost-effective than the original. Mileage dispersion is also substantial: approximately one quarter of the simulated draws are positive. In contrast, range-loss rate is the most uniformly penalized attribute: its standard deviation is small relative to the mean, and the distribution remains negative throughout nearly the entire sample. Together, these results suggest that consumer valuations of battery-related attributes are heterogeneous in both sign and magnitude. This motivates the segment-level analysis that follows.

Table 3
Mixed Logit Model Estimates in WTP Space (price unit: $10,000).

Parameter	Estimate	Std. Error	*p*-value		Summary of 10k Draws		
Mean parameters					1st Qu.	Median	3rd Qu.
λ (scale)	0.987	0.025	<0.001	***			
Mileage	-0.278	0.015	<0.001	***	-0.548	-0.277	-0.006
Range (Year 3)	1.107	0.045	<0.001	***	0.166	1.107	2.047
Range loss Rate (Year 3 to Year 8, in percentage)	-0.098	0.003	<0.001	***	-0.150	-0.098	-0.046
Battery Refurbishment: Pack Replace	-0.406	0.037	<0.001	***	-0.899	-0.407	0.085
Battery Refurbishment: Cell Replace	-0.448	0.037	<0.001	***	-0.899	-0.447	0.007
No choice (opt-out)	-5.130	0.110	<0.001	***			
Standard deviation							
SD: Mileage	-0.402	0.017	<0.001	***			
SD: Range (Year 3)	1.395	0.053	<0.001	***			
SD: Range loss Rate (Year 3 to Year 8, in percentage)	-0.077	0.003	<0.001	***			
SD: Battery Refurbishment: Pack Replace	0.730	0.060	<0.001	***			
SD: Battery Refurbishment: Cell Replace	-0.672	0.061	<0.001	***			
Observations							$3{,}072 \times 6 = 18{,}432$
Parameters							12
Log-Likelihood							-18,615.44
Null Log-Likelihood							-25,552.18
AIC							37,254.89
BIC							37,348.75
McFadden R^2							0.2715
Adj. McFadden R^2							0.2710

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1.

5.2. Results of Latent Class Choice Model

5.2.1. Model Selection

To determine the optimal number of latent classes, we estimated models with one through eight classes (Table 4). Model fit was evaluated using log-likelihood (LL), the ρ^2 measure relative to the null model, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), adjusted ρ^2, and entropy, where smaller AIC and BIC values and larger adjusted ρ^2 and entropy indicate better fit and class separation (Nylund et al., 2007). We selected a six-class solution on the basis of fit statistics and interpretability. The rate of BIC reduction slows substantially after six classes. Moreover, in the seven- and eight-class solutions, one class consistently accounted for fewer than 6% of the sample, reducing robustness for segment-level analysis.

Table 4
Latent Class Model Fit Index Comparison.

No. of Classes	Npar	LL	L^2	AIC	BIC	Adj.ρ^2	Entropy	Class proportions
1	11	-20,474	0	40,970	41,056	0.148	—	100.0%
2	33	-17,917	5,114	35,901	36,157	0.254	0.911	78.8% / 21.2%
3	55	-17,411	6,126	34,931	35,358	0.274	0.759	56.0% / 23.4% / 20.6%
4	77	-17,071	6,806	34,296	34,895	0.287	0.758	47.7% / 21.9% / 21.6% / 8.8%
5	99	-16,771	7,406	33,740	34,510	0.299	0.723	25.8% / 22.9% / 21.5% / 21.1% / 8.7%
6	121	-16,538	7,872	33,318	34,258	0.308	0.730	22.7% / 20.8% / 18.5% / 16.6% / 12.8% / 8.7%
7	143	-16,413	8,122	33,113	34,224	0.312	0.736	22.2% / 18.3% / 16.5% / 15.9% / 12.6% / 8.8% / 5.6%
8	165	-16,308	8,332	32,947	34,229	0.316	0.728	20.4% / 16.1% / 16.0% / 15.1% / 12.8% / 7.9% / 5.8% / 5.8%

[1] $L^2 = -2(\mathrm{LL}_{1c} - \mathrm{LL}_{\mathrm{model}})$.
[2] Adj.ρ^2 = adjusted rho-squared vs. observed shares (constants model).
[3] Entropy = $1 - H/H_{\max}$, where $H = -\sum \pi_{ik} \log(\pi_{ik})$. Values closer to 1 indicate cleaner class separation.

5.2.2. *WTP and Class Profiles*

Estimated model coefficients are presented in Table 5, with *Class 1* serving as the reference class. The full WTP estimates and class profile characteristics are reported in Table 6 in the Appendix, along with additional metadata on opt-out behavior and survey engagement (Table 7); readers can refer to these appendix tables for the exact numbers cited in the class descriptions below. Pairwise WTP difference tests across attribute segments are reported in Appendix Table 8. Figure 4 provides a visual summary of WTP estimates and segment profiles across classes. Class membership ranges from 8.7% to 22.7% of the sample. We describe each class below.

Class 1: Opt-Out Dominant, BEV-Skeptical Consumers (*n*=252, 8.7%).
Class 1 is defined primarily by systematic opt-out behavior: 75.4% of members selected the no-choice option in all six tasks, and an additional 19.4% did so in five of six. As a result, most WTP estimates for vehicle attributes are statistically indistinguishable from zero, with the exception of a moderate disutility for range loss in the 5–12% segment ($1,620* per percentage point) and significant penalties for both pack replacement ($6,350*) and cell replacement ($7,320*). These patterns suggest that respondents who systematically avoid the BEV market retain particular concerns about battery quality.

Class 1 represents consumers who are not yet viable participants in the used BEV market. The majority (70.4%) strongly disagree that they are likely to purchase a used BEV. They are the oldest class on average (47.1 years old), the least risk-tolerant (24.0% agree that they are prepared to take risks), and the least environmentally concerned (21.7% express concern about global climate change). Their household fleets are almost exclusively internal combustion engine vehicles (ICEVs), with 89.3% of households reporting no EVs and 93.3% driving an ICEV as their primary vehicle. A majority (68.2%) intend their next vehicle to be an SUV. EV-related knowledge is low (68.5% correctly answer the EV knowledge question and only 10.6% are aware of federal EV subsidies), while range anxiety is the highest of any class (87.2%). Attitudes toward refurbished batteries are notably negative: only 30.5% agree that they are environmentally beneficial, and just 9.1% disagree that they are functionally inferior. Infrastructure readiness is also limited: only 32.8% have home electrical outlet access, and only 24.7% report a neighbor who owns or leases a BEV or PHEV. Overall, this class is largely outside the addressable market for used BEVs in the near term.

Class 3: Range-Focused, EV-Engaged Consumers (*n*=485, 16.6%).
Class 3 contrasts sharply with Class 1, so we describe it next. It is distinguished by an exceptionally high WTP for range: $38,700*** per 100 miles for vehicles with 40–130 miles of range, falling to $20,820*** in 130–200 miles segment and rising again to $25,470*** above 200 miles. This convex pattern suggests that consumers in this class place a higher premium either on lower-range vehicles achieving a minimum viable range threshold (which may reflect their interest in a cost-effective secondary vehicle intended primarily for local trips), or on vehicles with very high ranges (which offer flexibility and reliability for long-distance travel). The pairwise tests confirm that the drop between the first and second range segments is statistically significant ($p = 0.003$; Appendix Table 8), indicating that Class 3 consumers genuinely distinguish minimum-viable-range vehicles from those with moderate range, rather than valuing each additional mile uniformly. This pattern is aligned with evidence that the marginal value of range is highest at low baseline range (Dimitropoulos et al., 2013; Hackbarth and Madlener, 2016). Consistent with this interpretation, members of this class report the highest average range for their current primary vehicle (321.3 miles), suggesting relatively demanding travel needs and greater sensitivity to vehicle range limitations. Their range loss aversion is moderate and uniformly significant across all degradation segments ($870***, $690***, $470***). The aversion to early-stage degradation (below 12%) is significantly stronger than to severe degradation (24%+) ($p = 0.031$), consistent with a threshold effect in which consumers treat a vehicle that has already lost a large share of its range as categorically impaired. Notably, WTP for refurbishment is negligible and statistically insignificant, suggesting that this class focuses more on current and future performance than on battery refurbishment history.

Class 3 represents the most EV-engaged segment in the sample. They hold the most favorable attitudes toward refurbished batteries (65.5% agree they are environmentally beneficial and 35.5% disagree that they are functionally inferior). Their households show the lowest rate of ICEV-only composition (74.1%), and they are the most likely class to prefer a car over an SUV (57.2%). They have the highest electrical outlet access (57.5%), along with the strongest EV knowledge (81.1%) and subsidy awareness (31.9%). They have the highest household income ($97,120) and the highest employment rate (71.5%), and are the most risk-tolerant of all classes (39.0% agree with a risk-taking characterization).

For this segment, auto dealers might filter out low-range inventory or offer clear range upgrade pathways. Their minimal disutility for refurbishment history, combined with high income, high EV knowledge, and positive attitudes toward refurbished batteries, suggests they are open to technically sound refurbishments provided the resulting vehicle meets their range threshold. Performance-oriented marketing that leads with current range and degradation trajectory, rather than battery service history, may resonate most with this segment.

Class 2: Battery Health Attentives (*n*=661, 22.7%).
Class 2, the largest class, is defined by battery SOH sensitivity. Range loss aversion is the highest across classes, with penalties of $2,310*** per percentage point in the 5–12% degradation segment, $2,330*** in the 12–24% segment, and $2,140*** in the 24%+ segment. Both refurbishment types also generate meaningful disutility ($2,740* for pack replacement; $3,760** for cell replacement). Their range WTP is consistent across the three piecewise segments ($4,570, $4,610, and $5,310**), indicating a steady marginal valuation of additional range without a strong threshold effect, though only the highest tier reaches statistical significance.

Members of Class 2 have an average household income of $91,340 and are relatively young (36.5 years). They demonstrate above-average EV knowledge (76.2%), moderate electrical outlet access (46.0%), and a high rate of EV-owning neighbors (43.5%). These well-informed consumers are buying battery reliability at the point of sale and protection against future performance loss more than range itself.

Their valuation structure is well suited to transparent, attribute-based pricing. Financing products that tie interest rates or warranty terms to battery SOH readings, or that offer extended coverage contingent on passing a minimum degradation threshold, could be particularly attractive to this segment. Their strong aversion to range loss also suggests that short-term ownership scenarios with rapid degradation are significantly discounted. Long-term degradation forecasts in the listing description could address this concern directly.

Class 5: Multi-Attribute Maximizers (*n*=606, 20.8%).
Class 5 exhibits statistically significant valuations across all vehicle attributes. Like Class 3, this class places a high premium on crossing the minimum viable range threshold, with still-substantial WTP across all range levels ($11,830***, $9,450***, $9,500***). The apparent decline across segments is directional but not statistically distinguishable (Appendix Table 8), suggesting a broadly uniform marginal valuation of range rather than a sharp threshold effect. Range loss aversion is significant across all three segments ($780, $760, $390 per percentage point). The decline from the low-degradation segment (<12%) to the severe segment (24%+) is statistically significant ($p = 0.032$), indicating that Class 5 consumers are notably more averse to early battery degradation than to incremental loss once degradation is already high. Refurbishment disutility is among the highest of the attribute-sensitive classes ($5,760*** for pack replacement; $6,070*** for cell replacement).

What further distinguishes Class 5 is its opt-out pattern: only 5.2% of members never opt out, while the plurality opt out two (29.0%) or three (27.2%) times out of six tasks. Class 5 members exhibit high range anxiety (83.4%) and are particularly averse to battery refurbishment (only 18.8% disagree that refurbished batteries are functionally inferior). They also report average household incomes ($86,920) and lower risk tolerance. These characteristics are consistent with their cautious evaluations across all vehicle attributes at once. The uniform magnitude and significance of their attribute benefits and penalties mean that no single attribute can easily offset large deficits elsewhere. Combined with their high opt-out frequency, this suggests that when a choice set presents no option that reaches a minimal quality threshold across all dimensions at once, respondents in this class choose to opt out rather than settle for the least-bad alternative.

Price discounts alone may be insufficient for this segment when a vehicle fails their quality bar on other dimensions. Product curation and clear multi-dimensional disclosure presenting range, degradation rate, and refurbishment status together could be a more effective approach for this group.

Class 6: Budget-constrained, Low-WTP Consumers (*n*=373, 12.8%).
Class 6 has the largest price coefficient in absolute value (−4.017***), more than twice the magnitude of the next-most price-sensitive class. This suggests that members of Class 6 are budget-constrained buyers. Range WTPs are modest but statistically significant at lower range segments ($2,810** for 40–130 miles; $1,760* for 130–200 miles), and range loss aversion is marginal ($170** and $170*** in the upper degradation segments). WTP for refurbishment is negligible and statistically insignificant. Battery service history is therefore largely immaterial in this group's purchasing calculus.

Class 6 members have the lowest average household income ($70,020) and the lowest next-vehicle budget ($19,660). They have the lowest homeownership rate (43.0%) and the highest share of renters (48.8%), the fewest household vehicles (1.9 on average), and the highest share of ICEV-only households (87.1%). Despite their strong price sensitivity, 91.2% of Class 6 members never select the no-choice option. They are interested in used vehicles, but price dominates battery SOH in their choices. For this class, a useful transparency tool would be a certified reliability standard at the point of sale that provides budget-constrained buyers a vehicle meets minimum quality requirements without requiring them to interpret complex battery health metrics. This would enable consumers to evaluate used BEVs with greater confidence and make purchase decisions that best align with their budget constraints and preferences.

Class 4: Low-Engagement, Attribute Non-Attendance Respondents (*n*=539, 18.5%). The most prominent feature of this class is its exceptionally low price coefficient in absolute value (−0.144**), which mechanically inflates all calculated WTP ratios. The class also shows the pattern that the choice-modeling literature associates with attribute non-attendance: one familiar attribute (mileage) carries a large, significant valuation ($9,950** per 10,000 miles) while most others are imprecisely estimated, and completion times are the shortest of any class except Class 1 (81 seconds across six tasks). Ignoring attributes is known to distort WTP estimates (Hensher et al., 2005), and a standard remedy is to let a latent class absorb non-attending behavior (Scarpa et al., 2009). We therefore retain Class 4 in estimation, where it serves this function, but flag its WTP values as unreliable and exclude them from cross-class comparisons.

Two subgroups likely coexist with the non-attenders. The class has a slightly higher share of extended-information respondents (52.7%), and the added battery cost and warranty detail may have amplified attention to refurbishment for some members, consistent with its extreme refurbishment penalties ($26,640** for pack replacement; $32,620** for cell replacement), though this evidence is at most weakly suggestive. The class also shows markers of EV experience: the highest household BEV ownership (13.5%), PHEV or HEV ownership (25.3%), and EV-owning neighbors (50.9%), the lowest range anxiety (67.0%), the highest risk-taking propensity (51.9%), and the highest next-vehicle budget ($29,290), suggesting some members regard battery-health attributes as low-stakes to their purchase decisions. These explanations cannot be separated with the present data, which is itself a reason to withhold substantive interpretation of this class.

Figure 4: WTP and class profile summary.

Table 5
LCCM model results

	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Vehicle Attributes						
No-Choice Option (opt-out)	-2.577 (1.344).	-9.819 (0.813)***	-1.968 (0.75)**	-3.491 (0.457)***	-2.513 (0.387)***	-13.847 (1.661)***
Mileage (10,000 miles)	-0.472 (0.248).	-0.3 (0.041)***	-0.31 (0.058)***	-0.143 (0.03)***	-0.22 (0.031)***	-0.229 (0.055)***
BEV Electric Range (Year 3): <130 miles	0.114 (1.326)	0.468 (0.281).	5.742 (0.417)***	0.177 (0.225)	1.426 (0.365)***	1.128 (0.399)**
BEV Electric Range (Year 3): 130-200 miles	1.884 (1.383)	0.472 (0.245).	3.089 (0.471)***	-0.145 (0.176)	1.14 (0.22)***	0.707 (0.357)*
BEV Electric Range (Year 3): 200+ miles	1.347 (0.61)*	0.545 (0.189)**	3.78 (0.733)***	0.198 (0.116).	1.145 (0.166)***	0.794 (0.507)
Range Loss Rate: <12%	-0.237 (0.072)***	-0.236 (0.029)***	-0.129 (0.024)***	0.004 (0.015)	-0.093 (0.017)***	-0.084 (0.03)**
Range Loss Rate: 12%-24%	-0.044 (0.063)	-0.239 (0.019)***	-0.102 (0.016)***	-0.018 (0.01).	-0.091 (0.013)***	-0.069 (0.02)***
Range Loss Rate: 24%+	0.035 (0.087)	-0.22 (0.03)***	-0.069 (0.019)***	-0.02 (0.01)*	-0.047 (0.019)*	-0.067 (0.02)***
Battery Refurbishment: Pack Replace	-0.932 (0.431)*	-0.281 (0.126)*	-0.08 (0.159)	-0.384 (0.105)***	-0.695 (0.106)***	0.041 (0.166)
Battery Refurbishment: Cell Replace	-1.074 (0.417)*	-0.385 (0.127)**	-0.099 (0.145)	-0.47 (0.102)***	-0.732 (0.11)***	0.109 (0.165)
Purchase Price (10,000 USD)	-1.467 (0.652)*	-1.025 (0.118)***	-1.484 (0.287)***	-0.144 (0.054)**	-1.206 (0.1)***	-4.017 (0.363)***
Active Indicators						
ASC	0	0.867 (0.5).	-0.272 (0.552)	2.641 (0.493)***	1.379 (0.539)*	1.28 (0.577)*
Perceived EV Range Anxiety: Agree	0	-0.021 (0.257)	-0.131 (0.272)	-0.734 (0.251)**	-0.118 (0.269)	-0.513 (0.283).
Risk Taking Propensity: Agree	0	0.692 (0.2)***	0.387 (0.214).	0.893 (0.204)***	0.042 (0.207)	0.001 (0.242)
Household Income (10,000 USD)	0	0.011 (0.016)	0.021 (0.018)	0.003 (0.017)	0 (0.018)	-0.048 (0.022)*
Electrical Outlet Access	0	0.391 (0.191)*	0.812 (0.205)***	0.525 (0.197)**	0.305 (0.195)	0.423 (0.223).
EV Battery Knowledge: Yes	0	0.146 (0.194)	0.307 (0.225)	-0.183 (0.202)	0.284 (0.197)	0.009 (0.223)
EV Battery Environmentally Positive: Agree	0	1.209 (0.189)***	1.295 (0.211)***	0.808 (0.196)***	0.829 (0.191)***	1.186 (0.217)***
EV Battery Functionally Negative: Disagree	0	0.987 (0.278)***	1.547 (0.284)***	0.822 (0.288)**	0.745 (0.29)*	1.169 (0.303)***
Current Primary Household Vehicle Fuel Type: ICEV	0	-0.629 (0.268)*	-0.592 (0.279)*	-1.227 (0.261)***	-0.596 (0.268)*	-0.015 (0.332)
Current Primary Vehicle Typical Range (miles)	0	-0.173 (0.091).	0.076 (0.098)	-0.325 (0.094)***	-0.113 (0.096)	-0.186 (0.101).
SUV/Crossover as the Next Vehicle	0	-0.203 (0.191)	-0.984 (0.273)***	-0.52 (0.198)**	-0.531 (0.191)**	-0.736 (0.222)***

Est. (SE)[Sig.]
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1.

5.3. Results of Opt-Out Analysis

The thematic network for the reasons for opt-out ($n = 209$) is plotted in Figure 5. The network visualization reveals 10 parent themes, which can be organized into four groups: general disinterest in BEVs, practical barriers (economic barriers, charging inconvenience, range anxiety, battery concerns), psychological factors (gas vehicle enthusiasm, EV distrust, used vehicle distrust, environmental concerns), and limited knowledge on BEVs. The codebook's residual "other" category received no mentions in the final sample.

The most prevalent reason is *general disinterest in BEVs* (n=54), expressed without reference to any concrete barrier (e.g., "I just don't want one"). *Economic barriers* ($n = 52$) group four sub-themes: the upfront purchase price, ongoing maintenance and repair expenses (in which battery replacement looms large), operation costs (e.g., electricity cost), and a generic "not worth it" value judgment. *Charging inconvenience* ($n = 51$) captures three sub-themes: insufficient access to public chargers (especially in rural or remote areas), lack of home charging access (often tied to apartment or rental living situations without a private outlet), and the time required to recharge relative to refueling a gasoline vehicle. *Range anxiety* ($n = 42$) encompasses two sub-themes: insufficient range for daily driving needs, and the fear of being stranded on long trips beyond a single charge. *Battery concerns* ($n = 37$) are anchored in three sub-themes: capacity degradation and lifespan loss, safety risks such as fire or crash damage to lithium-ion cells, and cold-weather performance loss.

Gas vehicle enthusiasm ($n = 37$) reflects a positive preference for internal combustion engine vehicles, split between an explicit love of gas engines (sound, range, refueling speed) and the absence of certain features in BEVs such as manual transmissions. *EV distrust* ($n = 28$) captures general skepticism of the technology, comprising a sub-theme of generic "no faith" statements and a related concern that the technology is still immature or not ready for mainstream adoption. *Used vehicle distrust* ($n = 13$) refers specifically to skepticism toward refurbished or used BEV batteries. *Environmental concerns* ($n = 8$) comprise a broader "green claim" skepticism (e.g., that BEVs are not actually better for the environment than gasoline vehicles) and concerns about end-of-life battery disposal and recycling. Finally, *Limited knowledge on BEVs* ($n = 3$) captures the smallest share who explicitly admitted insufficient familiarity to make an informed choice.

Following Krishna (2021), we further examine the co-occurrences among barriers, that is, the extent to which the same respondent simultaneously cites concerns from different parent themes. Among 209 responses, 62.7% ($n = 131$) cited a single barrier (i.e., one sub-theme), whereas 37.3% ($n = 78$) cited two or more (18.2% cited two, 12.9% cited three, and 6.2% cited four or more). Of the 78 multi-sub-theme responses, 74 cited sub-themes from different parent themes (e.g., *economic barriers* and *charging inconvenience*) and 4 cited multiple sub-themes within the same parent theme (e.g., *lack of public chargers* and *charging time*). To avoid over-reporting co-occurrences and over-crowding the figure, only pairs cited by five or more respondents are drawn as dashed lines in Figure 5, with the count labeled near each line; less frequent pairings are omitted. Theme counts report unique respondents, so sub-theme counts can sum to slightly more than their theme count when a respondent cited multiple sub-themes within the same theme.

The most frequent co-occurrences form a practical-infeasibility cluster linking charging and range: respondents who mention insufficient public chargers also mention long-trip range insufficiency ($n = 11$) or insufficient daily range ($n = 8$), and charging time co-occurs with insufficient daily range ($n = 7$). Battery-related concerns bridge into adjacent clusters: battery degradation co-occurs with used-vehicle distrust ($n = 6$), which suggests that distrust of used BEVs is partly rooted in battery replacement concerns, and with insufficient daily range ($n = 7$). Gas engine preference bridges to both practical and psychological barriers: love of gas engines co-occurs with insufficient public chargers ($n = 8$), general EV distrust ($n = 5$), and insufficient daily range ($n = 5$). Within a single theme, the only pairing above this threshold is daily and long-trip range insufficiency ($n = 7$). These overlaps suggest that addressing a single barrier (e.g., expanding public charging infrastructure) could, in principle, alleviate several adjacent concerns at once, consistent with the cascading-effect pattern reported by Krishna (2021).

Regarding the impact of the information treatment on opt-out, the opt-out rate in the total sample is 6.81% (=103/1,513) for the basic-information group and 6.80% (=106/1,559) for the extended-information group. A chi-square test finds no difference in the overall opt-out rate between treatment groups ($\chi^2 \approx 0$, $p = 1.00$).

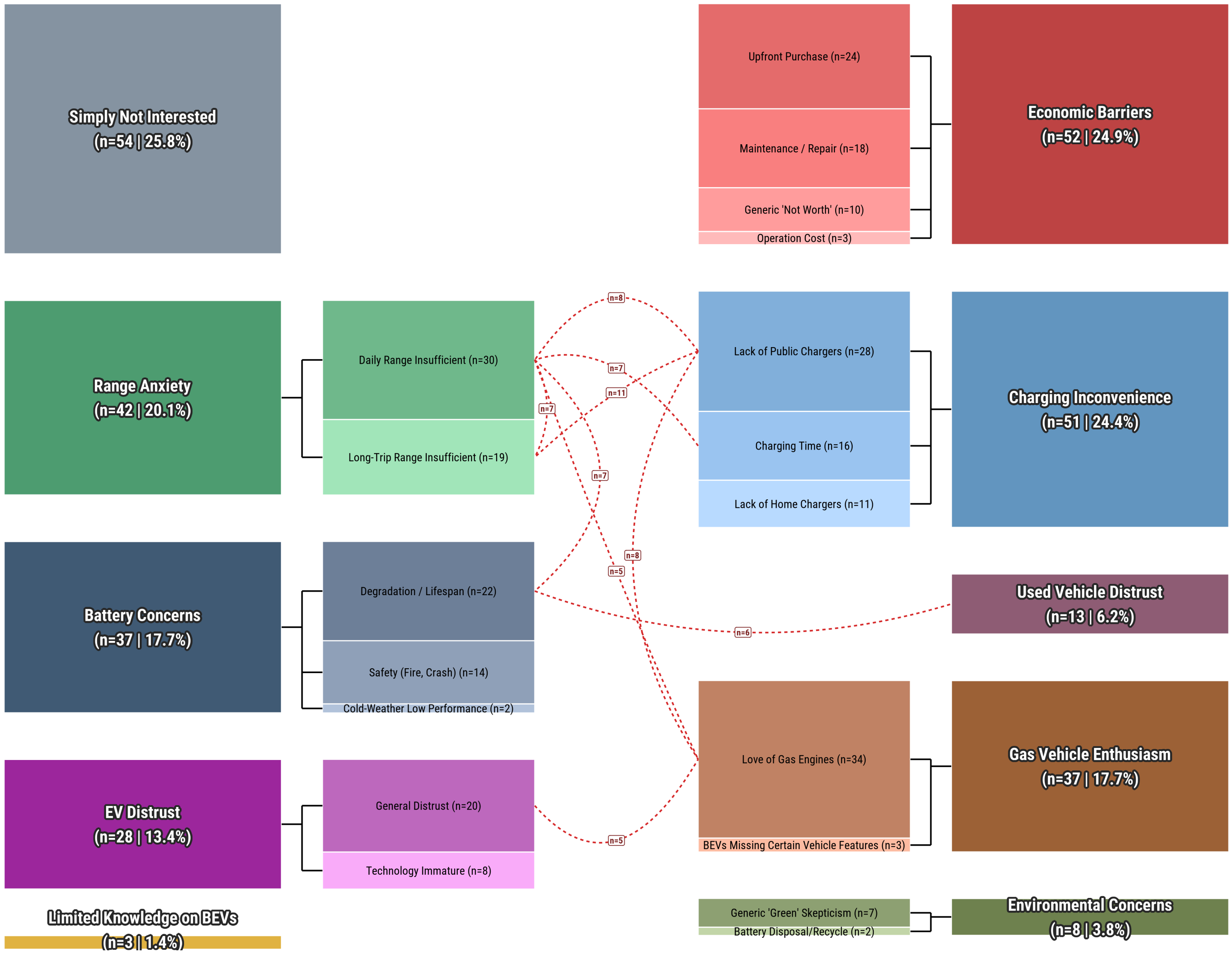

Figure 5: Thematic diagram and co-occurrences of opt-out reasons ($N = 209$).

Notes: 1. n counts unique respondents and their share of the 209 opt-out respondents. A respondent citing multiple sub-themes within a theme counts once toward the theme total, so sub-theme counts can sum to slightly more than their theme count; 2. Within each theme, sub-themes are shaded from darker to lighter in descending order of frequency. 3. Dashed lines mark pairs of sub-themes cited together by at least five respondents, with each label reporting the number of co-citing respondents; less frequent pairings are omitted.

5.4. Discussion

The DCE defined and displayed every battery-health attribute to every respondent. The WTP estimates therefore describe how consumers price these attributes when they can see them, which is the situation of a maturing used BEV market where battery-health information is standardized and visible at the point of sale, rather than today's market, where it is largely absent. The information treatment supports this interpretation: giving half the respondents additional detail on replacement costs and warranties changed neither market engagement nor the broad pattern of choices. Once the attributes themselves are visible, additional disclosure adds little. The value consumers express attaches more closely to the condition of the battery, not to the amount of information provided about it.

Consumer demand, however, is not a sufficient condition for battery-health certification in the short term. Certification at retail presupposes battery data at acquisition: where dealers source used BEVs through wholesale channels that carry no SOH information, they cannot price battery quality when they buy a vehicle, let alone certify it when they sell. Access to battery management system (BMS) data also remains gated by OEMs in the U.S., while the EU Battery Regulation requires read-only SOH access for independent operators from August 2024 (European Parliament and Council of the European Union, 2023), a contrast that shifts who can build certification in Europe but not in the U.S. Our estimates indicate that certification may be worth building. They do not indicate who can build it. In the U.S., OEM-backed certified pre-owned programs may be the most feasible near-term route, since the OEM controls both the battery data and the warranty, whereas independent certifiers face a data-access constraint that demand alone cannot lower.

In addition, the refurbishment WTP estimates warrant two caveats. First, the near-equality of the cell-level ($4,480) and pack-level ($4,060) penalties may reflect measurement rather than preference. Refurbished-battery used BEVs remain rare in retail inventory, so these estimates describe a future market. Second, even though the two service depths (cell-replaced versus pack-replaced) differ technically (Section 2), respondents encountered the distinction only through the attribute definitions, and buyers with no market experience of battery refurbishment may simply not differentiate them. The heterogeneity results from MXL also soften what the penalties mean in practice. Although the average buyer penalizes a refurbished battery, roughly a quarter of consumers do not mind one, and some even prefer it. Credible battery-health information would therefore work in two ways: it would let vehicles with documented healthy batteries earn higher prices from the buyers who care most about battery quality, and it would steer refurbished vehicles toward the price-focused buyers willing to accept them. The gain comes from matching each vehicle with the right buyer, not from raising every price.

6. Conclusion

The used BEV market is expanding, yet the persistent lack of information around battery health may continue to dampen consumer confidence and impede efficient pricing in the secondary market. This paper provides the first WTP estimates for battery-health attributes in the U.S. used BEV market, based on a national DCE with 3,072 respondents.

At the aggregate level, consumers value electric range, penalize mileage and battery degradation, and discount refurbishment history. The MXL estimates show that range is worth about $11,070 per 100 miles, degradation costs about $980 per point per year, and any refurbishment history costs $4,060+ on average while a quarter of buyers shrug at it. The preference heterogeneity is substantial. The LCCM suggests that consumers fall into six segments that differ markedly in their sensitivity to battery quality signals, their price responsiveness, and their market engagement. *Class 1* (8.7%) is defined by systematic opt-out behavior and represents consumers who are largely outside the addressable used BEV market, with low EV knowledge, high range anxiety, and predominantly ICEV households. *Class 2* (22.7%), the largest class, is characterized by the strongest and most uniform range loss aversion and meaningful disutility for both refurbishment types. These well-informed consumers prioritize battery reliability over range. *Class 3* (16.6%) is the most EV-engaged segment and places an exceptionally high premium on crossing a minimum viable range threshold. *Class 5* (20.8%) attends simultaneously to range adequacy, degradation trajectory, and refurbishment condition, and opts out selectively when no presented vehicle meets its standards across all dimensions at once. *Class 6* (12.8%) is the most price-sensitive class and is composed of budget-constrained buyers who are strongly committed to making a vehicle choice but largely indifferent to battery quality signals. *Class 4* (18.5%) is a low-engagement, attribute non-attendance class: its near-zero price sensitivity mechanically inflates WTP ratios, so we flag its estimates as unreliable and withhold substantive interpretation.

Beyond the choice models, the thematic analysis among respondents who opted out of all six tasks shows that market disengagement is not about battery health alone. The most common reasons are *general disinterest in BEVs*, *economic barriers*, and *charging inconvenience*, and over a third of these respondents cite more than one barrier. For the most disengaged consumers, better battery-health information addresses only one item on a longer list: their return to the market depends at least as much on charging access and affordability as on battery transparency.

This study has several limitations. The analysis relies on stated preferences from a DCE, which may not fully replicate the decision-making dynamics of real vehicle purchases, particularly for consumers with limited prior EV experience. Stated-preference estimates may also overstate market valuations. Our aggregate range estimate ($11,070/100mi) sits above new-vehicle benchmarks (2). The gap is consistent with the high marginal value of range in a short-range used fleet, and Class 3's range valuations in particular should be read with this caution. The large no-choice penalty (-$51,300) may imply a stronger compulsion to transact than real purchase behavior, a potential stated-preference artifact. Moreover, the sample is restricted to individuals who express an intention to purchase a used vehicle within two years, which excludes passive or uninterested populations. The LCCM assumes a finite number of discrete classes, so unobserved heterogeneity within each class remains. Future research can examine how these preferences evolve as used BEV supply increases and battery health reporting becomes more standardized, and whether incentive programs aimed at improving information quality in the secondary market translate into measurable shifts in demand.

7. Appendix

Table 6
WTP and Characteristics by Class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
WTP (*1000 USD) for Vehicle Attributes						
No-Choice Option (opt-out)	-$17.6.	-$95.8***	-$13.3**	-$242.4**	-$20.8***	-$34.5***
Mileage (10,000 miles)	-$3.2.	-$2.9***	-$2.1***	-$10.0**	-$1.8***	-$0.6***
BEV Range: 40-130 mi segment (per 100 miles)	$0.8	$4.6.	$38.7***	$12.3	$11.8***	$2.8**
BEV Range: 130-200 mi segment (per 100 miles)	$12.8	$4.6.	$20.8***	-$10.1	$9.5***	$1.8*
BEV Range: 200+ mi segment (per 100 miles)	$9.2*	$5.3**	$25.5***	$13.8.	$9.5***	$2.0
Range Loss Rate: 5-12% (per %)	-$1.6*	-$2.3***	-$0.9***	$0.3	-$0.8***	-$0.2**
Range Loss Rate: 12-24% (per %)	-$0.3	-$2.3***	-$0.7***	-$1.3.	-$0.8***	-$0.2***
Range Loss Rate: 24%+ (per %)	$0.2	-$2.1***	-$0.5***	-$1.4*	-$0.4*	-$0.2***
Battery Refurbishment: Pack Replace	-$6.4*	-$2.7*	-$0.5	-$26.6**	-$5.8***	$0.1
Battery Refurbishment: Cell Replace	-$7.3*	-$3.8**	-$0.7	-$32.6**	-$6.1***	$0.3
Active Indicators						
Risk-taking Propensity: Agree: no	76.0%	56.0%	61.0%	48.1%	71.3%	72.4%
Risk-taking Propensity: Agree: yes	24.0%	44.0%	39.0%	51.9%	28.7%	27.6%
Perceived EV Range Anxiety: Agree: no	12.8%	17.0%	19.4%	33.0%	16.6%	22.0%
Perceived EV Range Anxiety: Agree: yes	87.2%	83.0%	80.6%	67.0%	83.4%	78.0%
EV Battery Environmentally Positive: Agree: no	69.5%	37.4%	34.5%	46.4%	47.7%	39.6%
EV Battery Environmentally Positive: Agree: yes	30.5%	62.6%	65.5%	53.6%	52.3%	60.4%
EV Battery Functionally Negative: Disagree: no	90.9%	76.9%	64.5%	78.5%	81.2%	72.1%
EV Battery Functionally Negative: Disagree: yes	9.1%	23.1%	35.5%	21.5%	18.8%	27.9%
Household Income (1000 USD)	85.5	91.3	97.1	89.3	86.9	70.0
EV Knowledge: no	31.5%	23.8%	18.9%	28.4%	23.5%	30.7%
EV Knowledge: yes	68.5%	76.2%	81.1%	71.6%	76.5%	69.3%
Electrical Outlet Access: no	63.0%	47.8%	37.2%	42.3%	51.8%	52.3%
Electrical Outlet Access: not_sure	4.1%	6.3%	5.3%	5.8%	6.4%	7.8%
Electrical Outlet Access: yes	32.8%	46.0%	57.5%	51.9%	41.8%	39.8%
Primary Vehicle Typical Range (miles)	306.8	292.2	321.3	267.6	297.4	282.2
Household Vehicle Fuel Composition: has_bev	1.7%	7.5%	9.2%	13.5%	7.5%	3.4%
Household Vehicle Fuel Composition: has_phev_hev	7.6%	16.1%	16.6%	25.3%	13.3%	9.5%
Household Vehicle Fuel Composition: icev_only	89.3%	76.2%	74.1%	61.2%	79.0%	87.1%
Household Vehicle Fuel Composition: other	1.4%	0.1%	0.0%	0.0%	0.2%	0.0%
Next Vehicle Type: SUV: no	31.8%	38.9%	57.2%	48.3%	46.3%	53.3%
Next Vehicle Type: SUV: yes	68.2%	61.1%	42.8%	51.7%	53.7%	46.7%
Inactive Indicators						

Table 6
WTP and Characteristics by Class *(continued)*

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
EV Subsidy Knowledge: no	89.4%	73.6%	68.1%	70.9%	78.0%	79.1%
EV Subsidy Knowledge: yes	10.6%	26.4%	31.9%	29.1%	22.0%	20.9%
Neighbor Owns/Leases a BEV/PHEV: no	46.9%	29.6%	26.0%	28.5%	31.8%	33.9%
Neighbor Owns/Leases a BEV/PHEV: not_sure	28.3%	26.9%	26.8%	20.6%	30.3%	31.2%
Neighbor Owns/Leases a BEV/PHEV: yes	24.7%	43.5%	47.2%	50.9%	37.9%	34.9%
Age	47.1	36.5	37.5	37.4	40.9	37.7
Gender: female	67.3%	65.1%	52.6%	57.9%	63.1%	60.1%
Gender: male	31.3%	32.7%	44.9%	41.3%	34.4%	36.0%
Gender: other	1.4%	1.9%	2.5%	0.8%	2.5%	3.2%
Gender: prefer_not_answer	0.0%	0.3%	0.0%	0.1%	0.0%	0.7%
Ethnicity: hispanic	5.3%	11.6%	9.8%	15.5%	11.5%	11.6%
Ethnicity: non-hispanic	94.7%	88.4%	90.2%	84.5%	88.5%	88.4%
Race: african_american_only	7.1%	17.9%	14.4%	22.3%	11.4%	11.7%
Race: other	10.4%	15.6%	14.5%	18.5%	18.1%	15.9%
Race: white_only	82.5%	66.5%	71.2%	59.2%	70.5%	72.4%
Education Level: bachelor	31.9%	36.8%	38.4%	32.9%	36.1%	31.7%
Education Level: graduate	17.0%	20.2%	20.4%	20.4%	19.6%	17.9%
Education Level: high_school	18.6%	12.6%	11.9%	16.8%	11.4%	16.0%
Education Level: prefer_not_answer	0.0%	0.2%	0.0%	0.1%	0.5%	0.5%
Education Level: some_college	32.6%	30.2%	29.3%	29.8%	32.4%	33.9%
Student Status: non-student	92.4%	84.1%	84.6%	84.0%	86.0%	80.6%
Student Status: prefer_not_answer	0.7%	1.1%	0.8%	0.9%	1.8%	1.9%
Student Status: student	6.9%	14.8%	14.6%	15.1%	12.1%	17.5%
Employment Status: full_time	38.2%	47.2%	48.4%	46.9%	43.1%	38.6%
Employment Status: not_employed	36.5%	26.0%	27.7%	29.4%	33.1%	33.5%
Employment Status: part_time	24.6%	25.7%	23.1%	22.8%	21.9%	26.1%
Employment Status: prefer_not_answer	0.7%	1.1%	0.8%	0.9%	1.8%	1.9%
Household Size	2.8	3.1	2.9	3.3	2.9	2.8
Household Tenure: other	6.1%	6.1%	5.8%	4.7%	4.9%	7.2%
Household Tenure: own	58.3%	49.6%	54.4%	49.5%	52.5%	43.0%
Household Tenure: prefer_not_answer	1.6%	1.2%	1.8%	1.8%	1.6%	1.0%
Household Tenure: rent	34.1%	43.0%	38.1%	43.9%	41.0%	48.8%
Household Type: apart	17.2%	24.1%	21.6%	26.8%	21.9%	27.6%
Household Type: other	4.7%	3.2%	3.1%	3.2%	4.1%	3.9%
Household Type: sf_attached	9.2%	11.9%	9.2%	12.4%	10.6%	10.3%
Household Type: sf_detached	68.9%	60.9%	66.0%	57.6%	63.5%	58.2%
Household Vehicle Count	2.0	2.1	2.1	2.1	2.0	1.9
Primary Vehicle Fuel Type: bev	0.9%	3.1%	4.7%	6.1%	4.3%	1.4%
Primary Vehicle Fuel Type: icev	93.3%	85.3%	84.9%	72.0%	87.4%	91.6%
Primary Vehicle Fuel Type: phev_hev	5.8%	11.6%	10.4%	21.9%	8.3%	6.9%
Next Vehicle Budget (1000 USD)	24.3	27.8	26.9	29.3	25.2	19.7
Likelihood of buying new BEV: neutral	5.2%	10.7%	11.3%	14.8%	10.0%	10.0%
Likelihood of buying new BEV: somewhat_agree	5.3%	16.8%	16.7%	22.3%	13.1%	12.5%
Likelihood of buying new BEV: somewhat_disagree	10.6%	20.4%	22.2%	16.9%	16.4%	19.1%
Likelihood of buying new BEV: strongly_agree	1.7%	7.4%	5.8%	11.1%	4.6%	3.3%
Likelihood of buying new BEV: strongly_disagree	77.2%	44.6%	44.0%	34.9%	55.9%	55.2%
Likelihood of buying used BEV: neutral	5.5%	12.6%	13.2%	16.7%	12.3%	13.6%
Likelihood of buying used BEV: somewhat_agree	6.3%	26.2%	26.2%	26.7%	18.9%	23.4%
Likelihood of buying used BEV: somewhat_disagree	14.7%	20.5%	24.1%	18.0%	19.7%	22.0%
Likelihood of buying used BEV: strongly_agree	3.2%	10.5%	10.1%	11.1%	8.0%	7.5%
Likelihood of buying used BEV: strongly_disagree	70.4%	30.2%	26.4%	27.6%	41.2%	33.5%
Climate Concern: no	78.3%	56.5%	57.6%	62.4%	59.8%	58.4%
Climate Concern: yes	21.7%	43.5%	42.4%	37.6%	40.2%	41.6%

Table 7
Metadata Summary by Class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Metadata: Information Treatment						
Battery Information Treatment: no	51.6%	49.3%	50.7%	47.3%	48.3%	50.3%
Battery Information Treatment: yes	48.4%	50.7%	49.3%	52.7%	51.7%	49.7%
Metadata: Opt-out Count						
Opt-out: 0 times	0.0%	86.4%	85.7%	85.0%	5.2%	91.2%
Opt-out: 1 time	0.0%	12.0%	10.7%	12.2%	18.6%	7.4%
Opt-out: 2 times	0.0%	1.6%	2.9%	2.7%	29.0%	1.1%
Opt-out: 3 times	0.4%	0.1%	0.6%	0.1%	27.2%	0.3%
Opt-out: 4 times	4.8%	0.0%	0.0%	0.0%	15.3%	0.0%
Opt-out: 5 times	19.4%	0.0%	0.0%	0.0%	4.1%	0.0%
Opt-out: 6 times	75.4%	0.0%	0.0%	0.0%	0.6%	0.0%
Metadata: Survey Duration						
Battery DCE Section: Q1	22.0	25.0	27.0	22.0	27.0	25.0
Battery DCE Section: Q2	9.0	15.0	16.0	14.0	16.0	15.0
Battery DCE Section: Q3	7.0	14.0	15.0	12.0	14.0	13.0
Battery DCE Section: Q4	7.0	13.0	15.0	12.0	13.0	13.0
Battery DCE Section: Q5	6.0	12.0	13.0	11.0	12.0	12.0
Battery DCE Section: Q6	6.0	12.0	13.0	11.0	11.0	11.0
Battery DCE Section: Total	63.0	99.0	107.0	93.0	102.0	97.0
Full Survey	896.0	924.0	907.0	952.0	894.0	842.0

Table 8
Pairwise WTP differences across attribute levels, by latent class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
BEV Electric Range (Year 3, per 100 miles): Pairwise WTP Differences						
40–130 mi minus 130–200 mi	-$12,060	-$40	$17,880**	$22,370	$2,380	$1,050
40–130 mi minus 200+ mi	-$8,400	-$740	$13,230.	-$1,450	$2,330	$830
130–200 mi minus 200+ mi	$3,660	-$700	-$4,650	-$23,820	-$50	-$220
Range Loss Rate (per percentage point): Pairwise WTP Differences						
<12% minus 12–24%	-$1,320	$20	-$180	$1,520	-$20	-$40
<12% minus 24%+	-$1,860	-$170	-$400*	$1,640	-$390*	-$40
12–24% minus 24%+	-$540	-$190	-$220	$120	-$370	$0
Battery Refurbishment: Pairwise WTP Differences						
Pack replace minus Cell replace	$970	$1,020	$130	$5,980	$310	-$170

1 Each cell reports the WTP difference in dollars between two attribute levels; a positive value means the first level carries higher WTP (or lower disutility).

2 Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1.

Table 9
Monetized WTP estimates for vehicle and battery attributes in prior studies.

Study	Market and sample	Driving range	Charging	Operating cost	Other monetized attributes
Hidrue et al. (2011)	U.S.; potential BEV buyers	$35–$75 per additional mile	$425–$3,250 per hour reduction in charging time	About $2,706 for a $1/gallon-equivalent reduction	Acceleration: $2,600–$7,300 by improvement level. Pollution reduction: $1,900–$4,300+ by reduction level.
Helveston et al. (2015)	U.S. and China; new HEV, PHEV, and BEV alternatives	Not estimated separately; range enters through BEV75/100/150 alternatives. Implied U.S. value about $76 per mile (derived, not reported)	Fast charging: about $3,331 (U.S., PHEV; insignificant for BEV); about $7,567 (PHEV) and $6,428 (BEV) in China	About $1,600 (U.S.) and $3,000–$3,500 (China) per $0.01/mile reduction	BEV technology penalty: U.S. consumers value BEVs $10,000–$20,000 below a comparable conventional vehicle, depending on range. Acceleration: about $1,200 (U.S.) and $5,000 (China) per one-second improvement.
Forsythe et al. (2023)	U.S.; new car and SUV buyers (gasoline, HEV, PHEV, BEV)	$5,120 (cars) and $7,010 (SUVs) per 100 miles of BEV range	Fast-charging capability: $4,140 (cars) and $4,110 (SUVs)	$1,960 (cars) and $1,490 (SUVs) per $0.01/mile saved	Acceleration: $1,470 (cars) and $1,440 (SUVs) per one-second 0–60 mph reduction. BEV powertrain penalty: $4,160 (cars) and $8,700 (SUVs) relative to a comparable gasoline vehicle.
Dimitropoulos et al. (2013)	Meta-analysis of 33 BEV and AFV stated-preference studies	Mean $66–$75 per additional mile; marginal WTP declines as range increases	—	—	A 100-mile-range vehicle may need to be $13,000–$17,000 cheaper than a long-range conventional counterpart to be equally attractive.
Tanaka et al. (2014)	U.S. and Japan; BEV and PHEV choice	About $21.5 per 10 miles	—	$49.8 (U.S.) and $36.7 (Japan) per 1% fuel-cost reduction relative to gasoline	Fuel-station availability: $49.8 (U.S.) and $33.6 (Japan) per percentage point. Emissions reduction: $29.0 (U.S.) and $26.2 (Japan) per percentage point. Home plug-in installation: −$21.3 (U.S.) and −$16.9 (Japan) per $10 of installation cost.
Greene et al. (2018)	U.S.; pooled review of 52 vehicle-choice studies (2015 dollars; not BEV-specific)	Pooled mean $86 per mile across range variables	Pooled mean $2,195 per hour reduction in recharging time	Pooled mean $1,880 per $0.01/mile reduction	Acceleration: pooled mean $954 per one-second 0–60 mph reduction.
Ferguson et al. (2018)	Canada; latent class analysis (ICE-, HEV-, PHEV-, and BEV-oriented classes)	About $31/km (PHEV-oriented class) and $30/km (BEV-oriented class)	Public charging time saved: $531–$1,971 per hour across classes. Station availability: $317–$1,122 per step improvement (highest for the BEV-oriented class)	—	Battery warranty: $800–$3,153 per warranty-level improvement across classes, highest for HEV-oriented households, lowest for BEV-oriented households, who appear most comfortable with battery risk.

Notes: Values are nominal in each study's survey-year currency; cross-study comparison requires inflation adjustment. — indicates the attribute was not reported or not a focus of the study.

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References

Abbey, J. D., M. G. Meloy, V. D. R. Guide Jr., and S. Atalay (2015). "Remanufactured Products in Closed-Loop Supply Chains for Consumer Goods". In: *Production and Operations Management* 24.3, pp. 488–503. ISSN: 1937-5956. DOI: 10.1111/poms.12238. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/poms.12238 (visited on 07/06/2026).

Attia, P. M., A. Bills, F. Brosa Planella, P. Dechent, G. dos Reis, M. Dubarry, P. Gasper, R. Gilchrist, S. Greenbank, D. Howey, O. Liu, E. Khoo, Y. Preger, A. Soni, S. Sripad, A. G. Stefanopoulou, and V. Sulzer (June 2022). "Review—"Knees" in Lithium-Ion Battery Aging Trajectories". In: *Journal of The Electrochemical Society* 169.6, p. 060517. ISSN: 1945-7111. DOI: 10.1149/1945-7111/ac6d13. URL: https://dx.doi.org/10.1149/1945-7111/ac6d13 (visited on 07/03/2025).

Bashash, S., S. J. Moura, J. C. Forman, and H. K. Fathy (Jan. 1, 2011). "Plug-in Hybrid Electric Vehicle Charge Pattern Optimization for Energy Cost and Battery Longevity". In: *Journal of Power Sources* 196.1, pp. 541–549. ISSN: 0378-7753. DOI: 10.1016/j.jpowsour.2010.07.001. URL: https://www.sciencedirect.com/science/article/pii/S0378775310011390 (visited on 07/03/2025).

Bilfinger, P., M. Schreiber, P. Rosner, K. A. Gamra, J. Schöberl, C. Grosu, and M. Lienkamp (Aug. 21, 2025). *Why We Need a Standardized State of Health Definition for Electric Vehicle Battery Packs – a Proposal for Energy- and Capacity-Based Metrics*. DOI: 10.48550/arXiv.2508.15517. arXiv: 2508.15517 [eess.SY]. URL: http://arxiv.org/abs/2508.15517 (visited on 07/01/2026). Pre-published.

Boudway, I. (Dec. 16, 2020). "Batteries For Electric Cars Speed Toward a Tipping Point". In: *Bloomberg.com*. URL: https://www.bloomberg.com/news/articles/2020-12-16/electric-cars-are-about-to-be-as-cheap-as-gas-powered-models (visited on 07/03/2025).

Braun, V. and V. Clarke (Jan. 1, 2006). "Using Thematic Analysis in Psychology". In: *Qualitative Research in Psychology* 3.2, pp. 77–101. ISSN: 1478-0887. DOI: 10.1191/1478088706qp063oa. URL: https://doi.org/10.1191/1478088706qp063oa (visited on 06/28/2026).

Brückmann, G., M. Wicki, and T. Bernauer (Nov. 2021). "Is Resale Anxiety an Obstacle to Electric Vehicle Adoption? Results from a Survey Experiment in Switzerland". In: *Environmental Research Letters* 16.12, p. 124027. ISSN: 1748-9326. DOI: 10.1088/1748-9326/ac3531. URL: https://dx.doi.org/10.1088/1748-9326/ac3531 (visited on 06/30/2025).

Canals Casals, L., M. Rodríguez, C. Corchero, and R. E. Carrillo (Dec. 2019). "Evaluation of the End-of-Life of Electric Vehicle Batteries According to the State-of-Health". In: *World Electric Vehicle Journal* 10.4 (4), p. 63. ISSN: 2032-6653. DOI: 10.3390/wevj10040063. URL: https://www.mdpi.com/2032-6653/10/4/63 (visited on 07/02/2025).

Canepa, K., S. Hardman, and G. Tal (June 1, 2019). "An Early Look at Plug-in Electric Vehicle Adoption in Disadvantaged Communities in California". In: *Transport Policy* 78, pp. 19–30. ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2019.03.009. URL: https://www.sciencedirect.com/science/article/pii/S0967070X18303524 (visited on 07/01/2025).

Chen, C.-F., C. Fu, and Y.-C. Chen (Aug. 2023). "Exploring Tourist Preference for Mobility-as-a-Service (MaaS) – A Latent Class Choice Approach". In: *Transportation Research Part A: Policy and Practice* 174, p. 103750. ISSN: 09658564. DOI: 10.1016/j.tra.2023.103750. URL: https://linkinghub.elsevier.com/retrieve/pii/S0965856423001702 (visited on 05/18/2026).

Clarke, W. (2024). *Car Warranty Coverage on an Electric Car Battery*. US News & World Report. URL: https://cars.usnews.com/cars-trucks/advice/ev-battery-warranty (visited on 07/02/2025).

Cox Automotive Inc. (2025). *EV Market Monitor – February 2025*. URL: https://www.coxautoinc.com/market-insights/ev-market-monitor-february-2025/ (visited on 06/30/2025).

Dimitropoulos, A., P. Rietveld, and J. N. van Ommeren (Sept. 1, 2013). "Consumer Valuation of Changes in Driving Range: A Meta-Analysis". In: *Transportation Research Part A: Policy and Practice* 55, pp. 27–45. ISSN: 0965-8564. DOI: 10.1016/j.tra.2013.08.001. URL: https://www.sciencedirect.com/science/article/pii/S096585641300147X (visited on 07/07/2026).

Dnistran, I. (2024). *EV Batteries Can Outlast A Vehicle's Lifetime With Minimal Degradation, Study Finds*. InsideEVs. URL: https://insideevs.com/news/733987/ev-batteries-outlast-vehicle-degradation-study/ (visited on 07/02/2025).

European Parliament and Council of the European Union (July 28, 2023). *Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 Concerning Batteries and Waste Batteries*. URL: https://eur-lex.europa.eu/eli/reg/2023/1542/oj/eng.

Ferguson, M., M. Mohamed, C. D. Higgins, E. Abotalebi, and P. Kanaroglou (Jan. 2018). "How Open Are Canadian Households to Electric Vehicles? A National Latent Class Choice Analysis with Willingness-to-Pay and Metropolitan Characterization". In: *Transportation Research Part D: Transport and Environment* 58, pp. 208–224. ISSN: 13619209. DOI: 10.1016/j.trd.2017.12.006. URL: https://linkinghub.elsevier.com/retrieve/pii/S1361920916306058 (visited on 11/06/2025).

Fernandes, J. (2023). *US: Understanding Consumer Opinion about Used Electric Cars*. URL: https://business.yougov.com/content/47834-us-understanding-consumer-opinion-about-used-electric-cars (visited on 06/30/2025).

Forsythe, C. R., K. T. Gillingham, J. J. Michalek, and K. S. Whitefoot (June 6, 2023). "Technology Advancement Is Driving Electric Vehicle Adoption". In: *Proceedings of the National Academy of Sciences* 120.23, e2219396120. DOI: 10.1073/pnas.2219396120. URL: https://www.pnas.org/doi/10.1073/pnas.2219396120 (visited on 07/08/2026).

Foster, M., P. Isely, C. R. Standridge, and M. M. Hasan (Oct. 31, 2014). "Feasibility assessment of remanufacturing, repurposing, and recycling of end of vehicle application lithium-ion batteries". In: *Journal of Industrial Engineering and Management*. ISSN: 2013-0953. URL: https://raco.cat/index.php/JIEM/article/view/282032 (visited on 07/06/2026).

Gore, C., X. Ioganse, J. Kneifel, and J. Helveston (Nov. 18, 2025). "DCI: Consumer Valuation of Battery Health Information in Used Electric Vehicle Markets". In: *NIST*. URL: https://www.nist.gov/publications/dci-consumer-valuation-battery-health-information-used-electric-vehicle-markets (visited on 05/11/2026).

Greene, D. L., A. Hossain, J. Hofmann, G. Helfand, and R. Beach (2018). "Consumer Willingness to Pay for Vehicle Attributes: What Do We Know?" In: *Transportation Research Part A: Policy and Practice* 118, pp. 258–279.

Greene, W. H. and D. A. Hensher (2003). "A Latent Class Model for Discrete Choice Analysis: Contrasts with Mixed Logit". In: *Transportation Research Part B: Methodological* 37.8, pp. 681–698. URL: https://ideas.repec.org//a/eee/transb/v37y2003i8p681-698.html (visited on 05/18/2026).

Guzek, M., J. Jackowski, R. S. Jurecki, E. M. Szumska, P. Zdanowicz, and M. Żmuda (Jan. 2024). "Electric Vehicles—An Overview of Current Issues—Part 1—Environmental Impact, Source of Energy, Recycling, and Second Life of Battery". In: *Energies* 17.1 (1), p. 249. ISSN: 1996-1073. DOI: 10.3390/en17010249. URL: https://www.mdpi.com/1996-1073/17/1/249 (visited on 07/01/2025).

Hackbarth, A. and R. Madlener (Mar. 1, 2016). "Willingness-to-Pay for Alternative Fuel Vehicle Characteristics: A Stated Choice Study for Germany". In: *Transportation Research Part A: Policy and Practice* 85, pp. 89–111. ISSN: 0965-8564. DOI: 10.1016/j.tra.2015.12.005. URL: https://www.sciencedirect.com/science/article/pii/S0965856415002797 (visited on 07/07/2026).

Haghani, M., M. C. J. Bliemer, J. M. Rose, H. Oppewal, and E. Lancsar (Dec. 1, 2021). "Hypothetical Bias in Stated Choice Experiments: Part II. Conceptualisation of External Validity, Sources and Explanations of Bias and Effectiveness of Mitigation Methods". In: *Journal of Choice Modelling* 41, p. 100322. ISSN: 1755-5345. DOI: 10.1016/j.jocm.2021.100322. URL: https://www.sciencedirect.com/science/article/pii/S1755534521000555 (visited on 07/02/2026).

Hagman, J., S. Ritzén, J. J. Stier, and Y. Susilo (Mar. 1, 2016). "Total Cost of Ownership and Its Potential Implications for Battery Electric Vehicle Diffusion". In: *Research in Transportation Business & Management*. Innovations in Technologies for Sustainable Transport 18, pp. 11–17. ISSN: 2210-5395. DOI: 10.1016/j.rtbm.2016.01.003. URL: https://www.sciencedirect.com/science/article/pii/S2210539516000043 (visited on 07/01/2025).

Helveston, J. (Oct. 20, 2025). *cbcTools: Design and Analyze Choice-Based Conjoint Experiments*. Version 0.7.1. URL: https://cran.r-project.org/web/packages/cbcTools/index.html (visited on 10/22/2025).

Helveston, J. P. (Feb. 7, 2023). "Logitr: Fast Estimation of Multinomial and Mixed Logit Models with Preference Space and Willingness-to-Pay Space Utility Parameterizations". In: *Journal of Statistical Software* 105, pp. 1–37. ISSN: 1548-7660. DOI: 10.18637/jss.v105.i10. URL: https://doi.org/10.18637/jss.v105.i10 (visited on 06/24/2025).

Helveston, J. P., E. M. Feit, and J. J. Michalek (2018). "Pooling Stated and Revealed Preference Data in the Presence of RP Endogeneity". In: *Transportation Research Part B: Methodological* 109, pp. 70–89. URL: https://www.sciencedirect.com/science/article/pii/S0191261516304647 (visited on 06/05/2026).

Helveston, J. P., Y. Liu, E. M. Feit, E. Fuchs, E. Klampfl, and J. J. Michalek (Mar. 1, 2015). "Will Subsidies Drive Electric Vehicle Adoption? Measuring Consumer Preferences in the U.S. and China". In: *Transportation Research Part A: Policy and Practice* 73, pp. 96–112. ISSN: 0965-8564. DOI: 10.1016/j.tra.2015.01.002. URL: https://www.sciencedirect.com/science/article/pii/S0965856415000038 (visited on 07/01/2026).

Hensher, D. A., J. Rose, and W. H. Greene (May 1, 2005). "The Implications on Willingness to Pay of Respondents Ignoring Specific Attributes". In: *Transportation* 32.3, pp. 203–222. ISSN: 1572-9435. DOI: 10.1007/s11116-004-7613-8. URL: https://doi.org/10.1007/s11116-004-7613-8 (visited on 07/09/2026).

Hidrue, M. K., G. R. Parsons, W. Kempton, and M. P. Gardner (Sept. 1, 2011). "Willingness to Pay for Electric Vehicles and Their Attributes". In: *Resource and Energy Economics* 33.3, pp. 686–705. ISSN: 0928-7655. DOI: 10.1016/j.reseneeco.2011.02.002. URL: https://www.sciencedirect.com/science/article/pii/S0928765511000200 (visited on 07/01/2026).

Ioganse, X., C. Gore, J. Kneifel, S. Ranganath, and J. P. Helveston (Apr. 1, 2026). "Consumers ' Perceptions, Knowledge, and Adoption Patterns of Battery Electric Vehicles". In: *Transportation Research Part D: Transport and Environment* 153, p. 105194. ISSN: 1361-9209. DOI: 10.1016/j.trd.2025.105194. URL: https://www.sciencedirect.com/science/article/pii/S1361920925006042 (visited on 07/01/2026).

Ioganse, X., K. Wang, D. Bunch, G. Matson, and G. Circella (Feb. 1, 2023). "Deciphering the Factors Associated with Adoption of Alternative Fuel Vehicles in California: An Investigation of Latent Attitudes, Socio-Demographics, and Neighborhood Effects". In: *Transportation Research Part A: Policy and Practice* 168, p. 103535. ISSN: 0965-8564. DOI: 10.1016/j.tra.2022.10.012. URL: https://www.sciencedirect.com/science/article/pii/S0965856422002701 (visited on 06/10/2026).

Jiao, N. and S. Evans (Jan. 1, 2016). "Business Models for Sustainability: The Case of Second-life Electric Vehicle Batteries". In: *Procedia CIRP*. 13th Global Conference on Sustainable Manufacturing – Decoupling Growth from Resource Use 40, pp. 250–255. ISSN: 2212-8271. DOI: 10.1016/j.procir.2016.01.114. URL: https://www.sciencedirect.com/science/article/pii/S2212827116001293 (visited on 07/07/2025).

Kamakura, W. A. and G. J. Russell (Nov. 1, 1989). "A Probabilistic Choice Model for Market Segmentation and Elasticity Structure". In: *Journal of Marketing Research* 26.4, pp. 379–390. ISSN: 0022-2437. DOI: 10.1177/002224378902600401. URL: https://doi.org/10.1177/002224378902600401 (visited on 05/18/2026).

Kothari, S. (2024). *Battery Replacement Costs Are Poised To Plunge: 'Cheaper Than Fixing An Engine'*. InsideEVs. URL: https://insideevs.com/news/742022/battery-replacement-costs-fall-cheaper-than-fixing-engine/ (visited on 03/31/2025).

Krishna, G. (June 1, 2021). "Understanding and Identifying Barriers to Electric Vehicle Adoption through Thematic Analysis". In: *Transportation Research Interdisciplinary Perspectives* 10, p. 100364. ISSN: 2590-1982. DOI: 10.1016/j.trip.2021.100364. URL: https://www.sciencedirect.com/science/article/pii/S2590198221000713 (visited on 07/01/2025).

Letmathe, P. and M. Suares (Dec. 1, 2017). “A Consumer-Oriented Total Cost of Ownership Model for Different Vehicle Types in Germany”. In: *Transportation Research Part D: Transport and Environment* 57, pp. 314–335. ISSN: 1361-9209. DOI: 10.1016/j.trd.2017.09.007. URL: https://www.sciencedirect.com/science/article/pii/S1361920917304868 (visited on 07/07/2025).

Lim, M. K., H.-Y. Mak, and Y. Rong (Feb. 2015). “Toward Mass Adoption of Electric Vehicles: Impact of the Range and Resale Anxieties”. In: *Manufacturing & Service Operations Management* 17.1, pp. 101–119. ISSN: 1523-4614. DOI: 10.1287/msom.2014.0504. URL: https://pubsonline.informs.org/doi/10.1287/msom.2014.0504 (visited on 06/30/2025).

Loh, W. S. and R. B. Noland (Mar. 1, 2024). “Concerns Expressed by Used Electric Vehicle Owners Based on Surveying Social Media”. In: *Transportation Research Part D: Transport and Environment* 128, p. 104086. ISSN: 1361-9209. DOI: 10.1016/j.trd.2024.104086. URL: https://www.sciencedirect.com/science/article/pii/S1361920924000439 (visited on 07/01/2026).

Lohawala, N. and M. A. Rahman (May 1, 2026). “Do Determinants of EV Purchase Intent Vary across the Spectrum? Evidence from Bayesian Analysis of US Survey Data”. In: *Transportation Research Part A: Policy and Practice* 207, p. 104961. ISSN: 0965-8564. DOI: 10.1016/j.tra.2026.104961. URL: https://www.sciencedirect.com/science/article/pii/S0965856426001023 (visited on 07/01/2026).

Market Research Future (2025). *United States Used Car Market Trends & Growth By 2034*. URL: https://www.marketresearchfuture.com/reports/us-used-car-market-21771 (visited on 07/01/2025).

Mashrur, Sk. Md. and M. Mohamed (Feb. 1, 2025). “Uncovering Factors Affecting Consumers’ Decisions for Pre-Owned Electric Vehicles”. In: *Transportation Research Part D: Transport and Environment* 139, p. 104555. ISSN: 1361-9209. DOI: 10.1016/j.trd.2024.104555. URL: https://www.sciencedirect.com/science/article/pii/S1361920924005121 (visited on 11/18/2025).

McFadden, D. and K. Train (Sept. 2000). “Mixed MNL Models for Discrete Response”. In: *Journal of Applied Econometrics* 15.5, pp. 447–470. ISSN: 0883-7252, 1099-1255. DOI: 10.1002/1099-1255(200009/10)15:5<447::AID-JAE570>3.0.CO;2-1. URL: https://onlinelibrary.wiley.com/doi/10.1002/1099-1255(200009/10)15:5%3C447::AID-JAE570%3E3.0.CO;2-1 (visited on 06/05/2026).

Naseri, H., E. O. D. Waygood, Z. Patterson, and B. Wang (Sept. 1, 2024). “Who Is More Likely to Buy Electric Vehicles?” In: *Transport Policy* 155, pp. 15–28. ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2024.06.013. URL: https://www.sciencedirect.com/science/article/pii/S0967070X24001756 (visited on 07/01/2025).

Neubauer, J., A. Brooker, and E. Wood (July 1, 2012). “Sensitivity of Battery Electric Vehicle Economics to Drive Patterns, Vehicle Range, and Charge Strategies”. In: *Journal of Power Sources* 209, pp. 269–277. ISSN: 0378-7753. DOI: 10.1016/j.jpowsour.2012.02.107. URL: https://www.sciencedirect.com/science/article/pii/S0378775312005290 (visited on 07/03/2025).

Nylund, K. L., T. Asparouhov, and B. O. Muthén (Oct. 23, 2007). “Deciding on the Number of Classes in Latent Class Analysis and Growth Mixture Modeling: A Monte Carlo Simulation Study”. In: *Structural Equation Modeling: A Multidisciplinary Journal* 14.4, pp. 535–569. ISSN: 1070-5511. DOI: 10.1080/10705510701575396. URL: https://doi.org/10.1080/10705510701575396 (visited on 06/05/2026).

Park, J., K. Kim, S. Geum, J. Lee, H. Son, and S. Han (Mar. 23, 2026). *Battery Health Reporting Fails Independent Validation across Manufacturers*. DOI: 10.48550/arXiv.2603.21592. arXiv: 2603.21592 [eess.SP]. URL: http://arxiv.org/abs/2603.21592 (visited on 07/01/2026). Pre-published.

Pedrosa, G. and H. Nobre (Jan. 2018). “Second-Hand Electrical Vehicles: A First Look at the Secondary Market of Modern EVs”. In: *International Journal of Electric and Hybrid Vehicles* 10.3, pp. 236–252. ISSN: 1751-4088. DOI: 10.1504/IJEHV.2018.097378. URL: https://www.inderscienceonline.com/doi/abs/10.1504/IJEHV.2018.097378 (visited on 07/01/2025).

Scarpa, R., T. J. Gilbride, D. Campbell, and D. A. Hensher (June 1, 2009). “Modelling Attribute Non-Attendance in Choice Experiments for Rural Landscape Valuation”. In: *European Review of Agricultural Economics* 36.2, pp. 151–174. ISSN: 0165-1587. DOI: 10.1093/erae/jbp012. URL: https://doi.org/10.1093/erae/jbp012 (visited on 07/09/2026).

Shaikh, W. A., M. A. Kalwar, M. A. Khan, A. N. Wassan, M. H. Wadho, and M. F. Shahzad (2023). “A System Dynamics Costing Model for The Refurbishment of Electric Vehicle Batteries”. In: *Jordan Journal of Mechanical and Industrial Engineering* 17.01, pp. 15–32. ISSN: 1995-6665, 1995-6665. DOI: 10.59038/jjmie/170102. URL: http://jjmie.hu.edu.jo/vol17/vol17-1/02.htm (visited on 07/02/2025).

Sheykhfard, A., M. Azmoodeh, S. Das, and B. Kutela (Jan. 1, 2025). “Analyzing Purchase Intentions of Used Electric Vehicles through Consumer Experiences: A Structural Equation Modeling Approach”. In: *Transport Policy* 160, pp. 125–137. ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2024.10.038. URL: https://www.sciencedirect.com/science/article/pii/S0967070X24003305 (visited on 07/01/2025).

Skeete, J.-P., P. Wells, X. Dong, O. Heidrich, and G. Harper (Nov. 1, 2020). “Beyond the EVent Horizon: Battery Waste, Recycling, and Sustainability in the United Kingdom Electric Vehicle Transition”. In: *Energy Research & Social Science* 69, p. 101581. ISSN: 2214-6296. DOI: 10.1016/j.erss.2020.101581. URL: https://www.sciencedirect.com/science/article/pii/S2214629620301572 (visited on 07/07/2025).

Sonar, H., H. M. Belal, C. Foropon, R. Manatkar, and V. Sonwaney (July 20, 2023). “Examining the Causal Factors of the Electric Vehicle Adoption: A Pathway to Tackle Climate Change in Resource-Constrained Environment”. In: *Annals of Operations Research*. ISSN: 1572-9338. DOI: 10.1007/s10479-023-05519-8. URL: https://doi.org/10.1007/s10479-023-05519-8 (visited on 07/01/2025).

Tal, G., M. A. Nicholas, and T. S. Turrentine (2017). “First Look at the Plug-in Vehicle Secondary Market”. In: URL: https://escholarship.org/uc/item/22p191zs (visited on 07/01/2026).

Tanaka, M., T. Ida, K. Murakami, and L. Friedman (Dec. 1, 2014). “Consumers’ Willingness to Pay for Alternative Fuel Vehicles: A Comparative Discrete Choice Analysis between the US and Japan”. In: *Transportation Research Part A: Policy and Practice* 70, pp. 194–209. ISSN: 0965-8564. DOI: 10.1016/j.tra.2014.10.019. URL: https://www.sciencedirect.com/science/article/pii/S0965856414002614 (visited on 07/07/2026).

Tankou, A., G. Bieker, and D. Hall (2023). *Scaling up Reuse and Recycling of Electric Vehicle Batteries: Assessing Challenges and Policy Approaches*.

Tire Review Staff (Apr. 21, 2026). *EV Battery Repair vs Replacement Explained*. Tire Review Magazine. URL: https://www.tirereview.com/ev-battery-repair-vs-replacement/ (visited on 07/06/2026).

Train, K. E. (2009). *Discrete Choice Methods with Simulation*. 2nd ed. Cambridge: Cambridge University Press. ISBN: 978-0-521-76655-5. DOI: 10.1017/CBO9780511805271. URL: https://www.cambridge.org/core/books/discrete-choice-methods-with-simulation/49CABD00F3DDDA088A8FBFAAAD7E9546 (visited on 06/05/2026).

Webb, D., C. Gore, and J. Kneifel (Jan. 30, 2025). *Consumer Perspectives on Battery Electric Vehicles : An Analysis of Battery Information and Knowledge*. NIST TN 2328. Gaithersburg, MD: National Institute of Standards and Technology (U.S.), NIST TN 2328. DOI: 10.6028/NIST.TN.2328. URL: https://nvlpubs.nist.gov/nistpubs/TechnicalNotes/NIST.TN.2328.pdf (visited on 07/06/2026).

Witt, J. (2024). *Electric Car Battery Replacement Costs*. URL: https://www.recurrentauto.com/research/costs-ev-battery-replacement?utm_source=chatgpt.com (visited on 03/31/2025).

Yang, F., Y. Xie, Y. Deng, and C. Yuan (June 21, 2018). "Predictive Modeling of Battery Degradation and Greenhouse Gas Emissions from U.S. State-Level Electric Vehicle Operation". In: *Nature Communications* 9.1, p. 2429. ISSN: 2041-1723. DOI: 10.1038/s41467-018-04826-0. URL: https://www.nature.com/articles/s41467-018-04826-0 (visited on 07/03/2025).

Yuan, Q., W. Hao, H. Su, G. Bing, X. Gui, and A. Safikhani (2018). "Investigation on Range Anxiety and Safety Buffer of Battery Electric Vehicle Drivers". In: *Journal of Advanced Transportation* 2018.1, p. 8301209. ISSN: 2042-3195. DOI: 10.1155/2018/8301209. URL: https://onlinelibrary.wiley.com/doi/abs/10.1155/2018/8301209 (visited on 07/07/2025).

Zou, T., M. Khaloei, and D. MacKenzie (Dec. 1, 2020). "Effects of Charging Infrastructure Characteristics on Electric Vehicle Preferences of New and Used Car Buyers in the United States". In: *Transportation Research Record* 2674.12, pp. 165–175. ISSN: 0361-1981. DOI: 10.1177/0361198120952792. URL: https://doi.org/10.1177/0361198120952792 (visited on 07/01/2025).